JMLR: Workshop and Conference Proceedings 1–49, 2027 Conference on Health, Inference, and Learning (CHIL) 2027

Stated or Sampled? Post-Training Keeps the Base Model's Differential Diagnosis but Removes Its Variation Between Calls

Anonymous Authors
Anonymous Institution

Abstract

A language model's differential diagnosis can be read from the list it states in one answer or from the spread of its answers over repeated samples, the readout behind pass@k, self-consistency and sampling-based uncertainty. We measure how post-training changes both readouts on DDXPlus, whose synthetic patients carry a rule-based reference differential, with base/post-trained pairs from three open families at 7–32B (Qwen2.5 and Qwen3, every released stage of OLMo 3 7B and OLMo 3.1 32B, and Mistral-7B), zero- and few-shot prompts, and real rare-disease cases. Post-training keeps the base model's view of each patient but removes most of its variation between calls: a post-trained list is closer to its own base model's averaged list for the same patient than to that model's averages for other patients with the same diagnosis or to other families' base models, and it barely changes from call to call. Sampled answers collapse (1.3–2.4 distinct top answers per 32 samples at T=0.7, against 3.8–4.9), and with each model prompted at its best, post-training costs 13–30 points of pass@32; for one of two models tested, zero-shot sampling at a higher temperature recovers part of this. Decoded greedily with matched prompts, post-training significantly raises top-1 accuracy in at most two of six pairs, which suggests that much of its first-answer gain is the removal of sampling noise. One stated list improves robustly only for Qwen, where it contains the true pathology 12–24 points more often and gets about as close to the reference as five averaged base-model lists; post-trained lists recognize severe true diagnoses as well as other ones, but specific severe alternatives, most often acute pulmonary edema, are named less often, and asking explicitly for must-not-miss conditions recovers some of them, partly by lengthening the list and at the cost of more alarms. Majority voting stops helping and agreement between samples weakens as an error signal; combining it with the stated probability helps post-trained models only modestly. Qwen3-8B's thinking mode restores agreement as the better error signal and brings pass@32 to the base model's level, though not the breadth of its answers. Studies of diagnostic breadth should report which readout they use and whether the model reasons first.

Data and Code Availability. All data are public: DDXPlus (Fansi Tchango et al., 2022b) (synthetic patients) and Phenopacket Store v0.1.27 (Danis et al., 2025) (cases from published literature). All models are public checkpoints (Table 4). Prompts and an example vignette are in Appendix B; code and sampled outputs will be released with the camera-ready version.

Institutional Review Board (IRB). This work uses only public synthetic or previously published, de-identified data and involves no human subjects, so it did not require IRB review.

© 2027 A. Authors.

1. Introduction

Clinicians reason over a differential diagnosis, a ranked set of conditions consistent with the presentation, and the conditions they keep in view determine which tests are ordered and which dangerous possibilities are ruled out. A language model's differential can be read in two ways: from the list it states in one answer, or from the distribution of its answers over repeated samples. Much current practice uses the second. Pass@k, self-consistency voting (Wang et al., 2023) and sampling-based uncertainty (Farquhar et al., 2024) draw on repeated samples, and so does the evidence that reinforcement learning sharpens models, which win at pass@1 while their base models solve more problems given many samples (Yue et al., 2025; Wu et al., 2025; Oh et al., 2026). The two readouts agree for a model that samples from the distribution it states. Instruction-tuned models do not: they verbalize distributions they fail to sample from (Meister et al., 2025; Zhang et al., 2026; Jang et al., 2026), and one aligned answer can cover what many base-model samples spread out (Lake et al., 2025). We ask what this means for differential diagnosis, where every case has a reference differential and a true diagnosis, and what it does to the tools that read the differential from samples. This is a replication and extension in a clinical setting, not a new phenomenon.

The design holds the pretrained weights fixed and varies only the post-training: base/instruct pairs for Qwen2.5-7B, Qwen3-8B, Qwen2.5-32B and Mistral-7B; every released stage (SFT, DPO, RL) of OLMo 3 7B (Team Olmo et al., 2025) and OLMo 3.1 32B, plus a 7B checkpoint trained with RL directly from the base; three medical models fine-tuned from Qwen2.5-7B-Instruct; and Qwen3-8B with its thinking mode on. Each final pair is also run with worked examples in the prompt, so that base models answer in the requested format. Our measurements (Appendix A maps each to its evidence and checks):

- **Post-training keeps the base model's view and removes its variation.** A post-trained list tracks its own base model's averaged list for the same patient and barely changes between calls (§4.2).
- **Sampled answers collapse** in every family (13–30 points of pass@32 at each model's best prompt), while under greedy decoding with matched prompts post-training raises top-1 significantly in at most two of six pairs; the stage that narrows most depends on the recipe (§4.1).
- **The stated list improves robustly only for Qwen**, by naming the true pathology and the reference's conditions more often; severe conditions, also by two published must-not-miss lists, are named more often on average, but specific ones are lost (§4.2).
- **Sampling-based methods lose value**: voting stops helping and agreement weakens; combining it with the stated probability helps post-trained models modestly, and thinking mode restores agreement (§4.3). On real rare-disease cases the voting and agreement shifts replicate, but agreement stays the better signal (§4.4).

2. Related work

Post-training and the sampled distribution. RL with verifiable rewards improves pass@1 while base models win at large k on math and code (Yue et al., 2025; Wu et al.,

2025); a "sharpening tax" appears across 14 base/post-trained pairs on agentic tasks (Oh et al., 2026), though RL can expand the boundary in domains under-represented in pretraining (Cheng et al., 2025). Instruction tuning and RLHF reduce output diversity (Kirk et al., 2024; West and Potts, 2025; Jiang et al., 2025; Murthy et al., 2025), more so with scale for SFT (Springer et al., 2026), and concentrate next-token distributions so that outputs barely respond to decoding (Yang et al., 2026). Where diversity collapses depends on the pipeline: at DPO in OLMo and OLMo 2 (Peeperkorn et al., 2025) and in a stage ablation by Liu (2026), at each OLMo-2 stage for stated-versus-sampled distributions (Jang et al., 2026), and set by training data across Olmo 3's Think, Instruct and RL-Zero lineages (Karouzos et al., 2026). Base models given a few in-context examples approach aligned models (Lin et al., 2023), and theory and experiments cast post-training as sharpening the base model's own distribution toward its mode (Huang et al., 2024; Agarwal et al., 2025; Karan and Du, 2025), or as distilling its majority vote (Prasad et al., 2025). Large-k pass@k over few answer options is inflated by chance (Dragoi et al., 2025), and chat templates alone can narrow outputs (Yun et al., 2025).

Stated versus sampled distributions and uncertainty. Models asked to verbalize a distribution match it better than their samples do (Meister et al., 2025; Jang et al., 2026), and they struggle to draw from target distributions directly (Gu et al., 2026; Zhao et al., 2026). Verbalized sampling keeps most of the base model's diversity through Tülu 3's checkpoints (Zhang et al., 2026), and aligned answers aggregate content that base models spread across samples on open-ended questions with conflicting perspectives (Lake et al., 2025). Our stated readout is the verbalized top-k elicitation of Tian et al. (2023). For error detection, sample consistency usually beats verbalized confidence and hybrids beat both (Xiong et al., 2024; Lyu et al., 2024), including for diagnosis (Savage et al., 2025; Ren et al., 2026), confidence-weighted voting improves self-consistency (Taubenfeld et al., 2025), and agreement cannot flag errors a model repeats consistently (Tan et al., 2025); aligned models' samples can collapse enough to leave sampling-based uncertainty without signal (Liu, 2026), and cross-model disagreement (Hamidieh et al., 2026) or the base model (Luo et al., 2025) can supply what self-consistency misses.

Medical models and differential diagnosis. LLMs have been evaluated as differential-diagnosis aids on clinicopathological cases (Kanjee et al., 2023; McDuff et al., 2025) and on Phenopacket Store, where they trail the phenotype-driven tool Exomiser (Reese et al., 2026; Chimirri et al., 2025). Medical post-training includes verifier-guided SFT and PPO (Chen et al., 2025), rule-based RL (Liu et al., 2025) and reasoning distillation (Huang et al., 2026); RL and reward models tend to favor confident answers (Leng et al., 2025; Damani et al., 2026; Mei et al., 2026). DDXPlus scores predicted against reference differentials as sets (Fansi Tchango et al., 2022a); H-DDx proposes hierarchical F1 (Lim et al., 2025), MEDDxAgent studies interactive diagnosis (Rose et al., 2025), and Puri et al. (2026) show on DDXPlus with Qwen3-8B that single-answer RL collapses sampled answers and train models to output several answers per completion. Relative to Jang et al. (2026), who compare stated and sampled distributions on synthetic and survey targets with two of our pairs, Puri et al. (2026) and Karouzos et al. (2026), we add a reference distribution with a true diagnosis, a direct test that post-trained lists reproduce the base model's per-case

view, prompt, temperature and decoding controls, and the consequences for voting, error signals and severe conditions.

3. Setup

Tasks. **DDXPlus** (Fansi Tchango et al., 2022b): synthetic patients from a rule-based medical knowledge base, each with a true pathology and a reference differential, the engine's posterior over 49 conditions. We use 980 test patients (20 per condition) rendered by template as questionnaire-style vignettes without exam or test results (example in Appendix B). The prompt lists the 49 conditions and asks for the five most likely, with probabilities, as JSON, so this is a closed-set task. **Phenopacket Store** (Danis et al., 2025): 1,401 published rare-disease cases covering 749 diseases (at most two per disease), rendered as phenotype vignettes with an open-ended differential; names are graded against the case's diagnosis via Mondo as exact, related (a specific ancestor group or subtype) or wrong (Appendix B).

Models and sampling. Qwen2.5-7B, Qwen3-8B, Qwen2.5-32B and Mistral-7B-v0.3, each with its official post-trained version; OLMo-3-1025-7B and OLMo-3-1125-32B with the SFT, DPO and final RLVR (Instruct) checkpoints of OLMo 3 7B and OLMo 3.1 32B, plus OLMo-3-7B-RL-Zero-General, trained with RL from the 7B base; and AlphaMed, HuatuoGPT-o1 and m1, fine-tuned from Qwen2.5-7B-Instruct (Table 4). We draw 32 samples per case with vLLM at T=0.7 (top-p 0.95) and, for the 7–8B models, T=1 (top-p 1), each model with its own chat template (base models without one get a post-trained sibling's) and Qwen3's thinking mode off unless stated. A sample's *top answer* is valid if it is one of the 49 conditions and its *list* if it names at least one; invalid answers count as wrong. With this zero-shot prompt base models are often invalid (up to 47% of samples for OLMo and 76% for Mistral), so every final pair is re-run with three worked examples from the training split (Appendix F.1), on which every model is valid in over 99% of samples.

Readouts. The *sampled* readout is the distribution of top answers over a case's samples: pass@k (Chen et al., 2021), distinct top answers, the reference mass they cover (coverage@32), and total-variation distance (TV) of their empirical distribution to the reference. The *stated* readout is one sample's list, with its probabilities renormalized over the conditions it names (uniform if missing), averaged per case over valid samples: whether it contains the true pathology among its top k (recall@k), its TV to the reference (also with the true pathology removed from both), the reference mass it names, its *conditional* TV (to the reference restricted to the named conditions and renormalized, a scale-free measure of how it allocates probability), DDXPlus's set F1 against the reference's support (DDF1), and whether it names each severe condition (DDXPlus severity 1–2, the dataset's label) that the reference weights at least 0.1. The *single-call answer* is the top answer of a model's first valid sample.

Statistics. DDXPlus has 20 cases per true pathology, so intervals are 95% percentile intervals from a bootstrap that resamples the 49 pathologies with all their cases (10,000 resamples) unless stated; p-values use the add-one correction. The 112 headline tests (14 metrics fixed in advance, eight zero-shot settings) are primary and Holm-corrected together; every control, stage, few-shot and per-condition analysis is secondary, corrected within its

own family (Appendix J), and read as a check on them. Case-level intervals in the appendix are up to three times narrower, and some appendix tables come from an earlier scorer whose values differ by up to 0.004. AUROCs use average ranks for ties. With five to six pipelines, results describe these checkpoints rather than post-training in general.

4. Results

4.1. Post-training collapses the sampled answers

Table 1 compares each base model with its post-trained versions. At 7–8B the post-trained Qwen model is far better on its first answer (+12.6 to +27.5 points), but its pass@k curve crosses the base model's at k≈4–8, and at k=32 the base model leads by 9.7–14.6 points (Figure 2 in Appendix D.1). At 32B the zero-shot losses are smaller and uncertain (Qwen2.5-32B −4.9 [−10.3, +0.5], OLMo 3.1 32B −4.0 [−6.8, −1.2], neither surviving correction). Base models answer the zero-shot prompt invalidly far more often, and some post-trained models do worse with worked examples (Qwen2.5-7B-Instruct 0.598 → 0.535), so we also compare each base model with worked examples against its post-trained model at its better prompt: post-training then costs 13.2–29.9 points of pass@32 in all six final pairs, including both 32B pairs and Mistral-7B, also with example lists taken from another model instead of the reference (Appendices F.1–F.5). Part of this is sharper sampling. Zero-shot, Qwen2.5-7B-Instruct at T=1.3 matches its base model's pass@32 at T=0.7 (0.704 against 0.695), though not at T=1 (0.766), at a cost of 4.2 points of pass@1, whereas Qwen3-8B stays below its base model even at T=1.6 (0.684 against 0.732; Appendix F.2). The first-answer gain has the same source: decoded greedily, post-training significantly raises top-1 in at most two of six pairs, whether both arms see the same worked examples (Qwen3-8B and Mistral) or neither sees a reference list (Qwen3-8B), and lowers it significantly in up to two (Appendix F.3). Across temperatures the base and post-trained frontiers meet for Qwen2.5-7B, and the base model dominates for OLMo 3 7B (Table 19). Post-trained models give 1.3–2.4 distinct top answers per 32 samples at T=0.7, against 3.8–4.9. Coverage@32 rewards spreading answers (a uniform guesser covers 0.48, more than any model), but TV does not: samplers that always give each case's reference mode or its true pathology score 0.694 and 0.714, better than a guesser that ignores the patient (0.774; Table 2). Base models' samples lie at 0.73–0.76, and post-trained samples are farther than the one-answer oracles in every setting and farther than the patient-blind guesser for three of the five final models (0.79–0.85). Which stage narrows the samples most depends on the recipe: in OLMo 3 7B the RLVR step costs 19.1 [13.8, 24.6] points of pass@32 against the DPO model, in OLMo 3.1 32B the SFT step costs 12.1 [6.8, 18.0] and RLVR 2.8 [0.5, 5.3], and RL-Zero, trained with RL from the 7B base, loses 13.3 [7.3, 19.7] points at T=0.7 but none detectably at T=1 (Appendices D.4, D.5). Qwen3-8B with thinking on reaches its base model's pass@32 (0.815 against 0.732 zero-shot and 0.809 with worked examples) with samples as close to the reference as the base model's, though still narrow (1.76 distinct answers; Appendix G.1).

4.2. Post-training keeps the base model's view and removes its variation

What is a post-trained list? For each patient we compare each post-trained list with the mean of its base model's lists for the same patient (Table 24). In all 15 settings the post-

Comparison	Model	Sampled top answers pass@1	pass@32	Distinct	Cov@32	Drawn from stated lists pass@32	Distinct
Qwen3-8B, T=0.7	base	0.388	**0.732**	**3.77**	**0.323**	0.780	7.79
	post-trained	**0.559**	0.607	1.27	0.217	**0.811**	4.96
Qwen3-8B, T=1	base	0.282	**0.789**	**6.60**	**0.407**	**0.832**	9.82
	post-trained	**0.556**	0.643	1.53	0.236	0.820	5.40
Qwen2.5-7B, T=0.7	base	0.352	**0.695**	**4.91**	**0.335**	0.772	10.68
	instruct	**0.478**	0.598	1.82	0.239	**0.802**	6.81
Qwen2.5-7B, T=1	base	0.239	**0.766**	**8.23**	**0.422**	0.785	12.44
	instruct	**0.464**	0.663	2.65	0.290	**0.827**	7.76
Qwen2.5-32B, T=0.7	base	0.500	**0.795**	**3.78**	**0.352**	0.850	8.79
	instruct	**0.604**	0.746	1.68	0.276	**0.880**	6.49
OLMo 3 7B, T=0.7	base	0.178	**0.705**	**4.93**	**0.351**	**0.719**	8.66
	SFT+DPO+RLVR	0.265	0.422	2.37	0.195	0.577	5.65
	RL-Zero (from base)	**0.295**	0.572	4.06	0.286	0.701	9.66
OLMo 3.1 32B, T=0.7	base	0.326	**0.796**	**4.32**	**0.347**	0.839	8.11
	SFT+DPO+RLVR	**0.560**	0.756	2.15	0.283	**0.853**	6.75
Chance	uniform guesser	0.020	0.48	23.7	0.48		
	reference-prior guesser	—	0.44	21.9	0.56		

Table 1: Sampled readout on DDXPlus (980 cases, 32 samples). Distinct: distinct top answers per case; Cov@32: reference mass they cover. *Drawn from stated lists*: each sample's top answer replaced by a draw from its own stated list. Bold: better model of each pair (OLMo: of the block; the SFT and DPO stages are in Tables 8 and 9). Invalid answers count as wrong, which mostly penalizes base models. Chance: guessers that ignore the patient, drawing uniformly from the 49 conditions or from the reference differential averaged over test cases.

trained list is closer to that average than to the base model's averages for other patients with the same true pathology (TV −0.028 to −0.076) and closer than to the same-patient averages of the other family's base models (by 0.03–0.14 zero-shot); within a family the distinction blurs, since Qwen2.5-7B-Instruct is as close to the other Qwen base models and OLMo-3-7B-Instruct is closer to OLMo-3-1125-32B's. It is about as close to the base average as a single base-model list at T=0.7 (Qwen3-8B 0.368 against 0.372) and closer at T=1, but post-trained lists barely vary between calls (TV 0.10–0.33 to the mean of the model's other lists, against 0.37–0.69 for base lists). Post-training thus keeps the base model's case-specific view and removes its variation between calls. OLMo 3 7B is the exception: its lists drift from its base model's view (0.628 against 0.493).

Whether this improves the list depends on the model (Table 2). For every Qwen model and OLMo 3.1 32B, the post-trained list contains the true pathology among its top five 12–24 points more often (e.g. Qwen3-8B 0.67 → 0.80), is closer to the reference (TV −0.043 to −0.088; −0.018 to −0.056 without the true pathology, Qwen3-8B at T=0.7 not surviving correction), names more reference mass, and allocates probability among the named conditions more like the reference (conditional TV −0.025 to −0.078). OLMo 3 7B's lists do not improve (recall@5 0.47 → 0.49), and without the true pathology they are farther (+0.034 at T=0.7). The same gains hold on complete five-item lists only (recall@5 +0.12 to +0.16). With worked examples the recall and TV gains persist for Qwen3-8B

and Qwen2.5-7B but recall does not for Qwen2.5-32B, OLMo 3.1 32B keeps only a higher reference mass, and Mistral gains with either set of examples (TV −0.036 and −0.059): the stated-list gain is consistent for the smaller Qwen models, holds for Mistral with worked examples, and is weak or absent for OLMo. When the true pathology is itself severe (340 cases), post-trained lists contain it among their top five 9–18 points more often zero-shot, as for other pathologies. Averaging calls closes most of the gap: the mean of five base-model lists is about as close to the reference as one post-trained list (Qwen3-8B 0.699 against 0.700, OLMo 3.1 32B 0.691 against 0.701; diamonds in Figure 1a). With five lists on both sides, post-training is closer in four of eleven direct-answer settings (all Qwen), not detectably different in four and farther in three (OLMo; Appendix G.2). Qwen3-8B with thinking on goes further: one of its lists (TV 0.670) beats five averaged base lists.

Severe conditions. Post-trained lists name the severe conditions that the reference weights at least 0.1 more often in every zero-shot setting (Qwen +9.0 to +13.8 points, OLMo 3 7B +6.0 and +8.7, the first not surviving correction; OLMo 3.1 32B +2.2, n.s.), and the mean inclusion over all 17 severe conditions rises significantly in seven of eight. Yet one list names only 0.32–0.47 of the warranted ones, against 0.975 for the reference's top five, and specific conditions are lost (Table 11): acute pulmonary edema is the largest loss in four of the five final pairs (e.g. Qwen2.5-32B 0.63 → 0.44, OLMo 3 7B 0.56 → 0.31). With worked examples the mean inclusion falls for Qwen2.5-32B and OLMo 3 7B, and five averaged post-trained lists name fewer warranted severe conditions than five averaged base lists in six of eleven settings (by 0.06–0.14). The Qwen models shift toward acute coronary syndromes and pulmonary embolism, which emergency medicine rules out first, but these vignettes lack the ECG and biomarkers that separate the coronary syndromes, so the shifts track the reference engine rather than clinical correctness. With must-not-miss conditions taken instead from diagnostic-harm data and the chest-pain guideline (Newman-Toker et al., 2024; Gulati et al., 2021), post-trained lists name 9–24 points more of them in seven of eight zero-shot settings, and only OLMo, Mistral and the few-shot Qwen2.5 models lose specific ones (Appendix F.4). Asked explicitly for up to three dangerous conditions to rule out, post-trained lists name 6–15 points more of the warranted severe conditions and name acute pulmonary edema more often for five of six models, at a cost of 1–2 more severe conditions per list that the reference does not warrant, some of them plausible rule-outs such as pulmonary embolism (Table 20); against two combined standard lists, which name one to two fewer conditions, 0.02–0.11 of the gain remains.

4.3. Consequences for sampling-based methods

Aggregation. Majority voting over 32 samples (self-consistency) raises base models' accuracy by 3.4–8.4 points at T=0.7 but adds at most 1.9 for the final post-trained models (Qwen2.5-32B-Instruct, $[0.8, 3.1]$; the others n.s.; Table 26 in Appendix G.3). Models that reason before answering still gain (HuatuoGPT-o1 +4.8, m1 +3.8, Qwen3-8B with a CoT prompt +4.5), and so does Qwen2.5-7B-Instruct sampled at T=1.3 (+5.6). For a top-three differential, the top three of one list contains the true pathology more often than the three most frequent of 32 samples for the post-trained Qwen models (+5.6 to +13.7 points), while for every base model the samples win by 7.3–13.6.

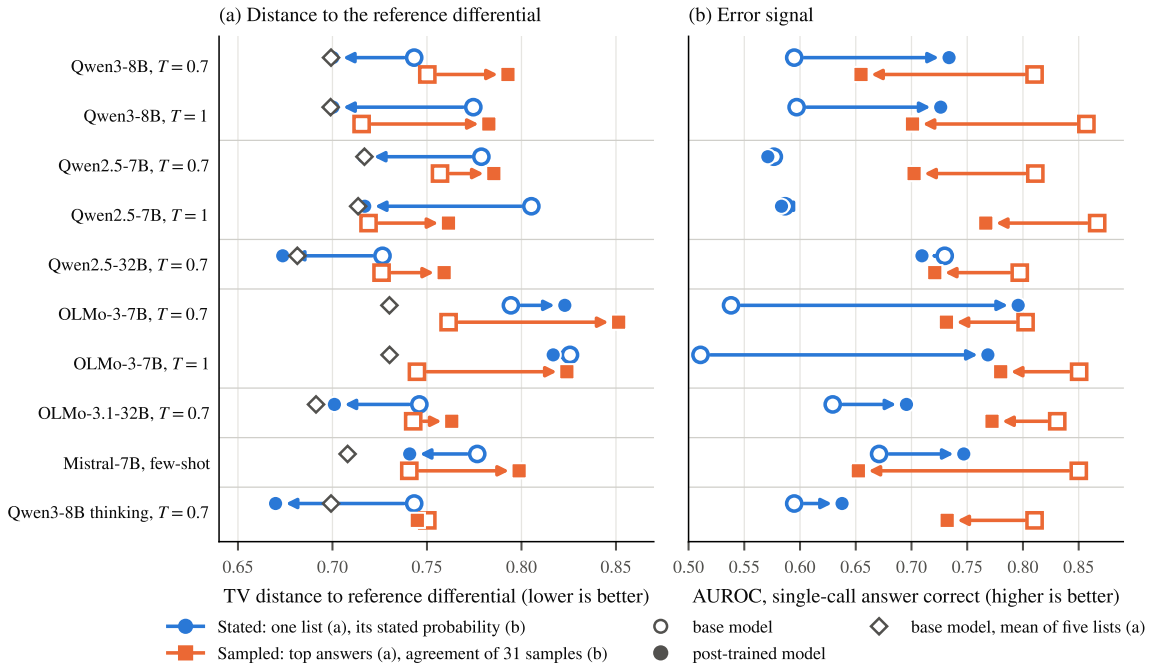


Figure 1: Base (hollow) → post-trained (filled) on DDXPlus (980 cases; OLMo: base → final Instruct; Mistral-7B with the few-shot prompt; the last row compares Qwen3-8B-Base with Qwen3-8B with thinking on). (a) TV to the reference differential of one stated list (blue circles) and of the sampled top answers (orange squares); grey diamonds: the mean of the base model's first five lists. (b) AUROC for the single-call answer being correct, of its stated probability (blue) and of agreement across 31 further samples (orange).

Uncertainty. We score each signal by how well it ranks the correctness of the single-call answer (AUROC); the stated probability needs no extra calls, *agreement* (the share of 31 further samples with the same top answer) needs them. For base models agreement beats the stated probability by 0.07–0.26 (not significantly for Qwen2.5-32B). Post-training weakens it in every family (Qwen3-8B 0.810 → 0.655, Qwen2.5-7B 0.811 → 0.702, Qwen2.5-32B 0.797 → 0.721, OLMo 3 7B 0.802 → 0.731, OLMo 3.1 32B 0.831 → 0.772; Figure 1b), significantly for Qwen but not OLMo, consistent with Liu (2026); since collapsed samples often tie, which pushes agreement's AUROC toward 0.5, this largely restates the collapse. The stated probability then beats agreement for Qwen3-8B (+0.079 [0.011, 0.143]), while agreement stays better for Qwen2.5-7B-Instruct and OLMo-3.1-32B-Instruct (−0.131, −0.077); with worked examples the stated probability wins for Qwen3-8B, Qwen2.5-7B-Instruct and Mistral-Instruct (+0.095 to +0.130) and ties for the others: the better signal depends on model and prompt. Thinking returns Qwen3-8B to the base pattern (agreement better by 0.094 [0.027, 0.170]), and so does sampling Qwen2.5-7B-Instruct at T=1.3 (0.794 against 0.602).

Comparison (T=0.7)	Model	TV to reference↓ One list	Samples	List w/o gold	Reference mass named	Cond. TV↓	DDF1	Recall @5	Severe named
Qwen3-8B	base	0.743	**0.750**	0.833	0.334	0.492	0.276	0.666	0.347
	post-trained	**0.700**	0.793	**0.815**	**0.396**	**0.431**	**0.318**	**0.802**	**0.437**
Qwen2.5-7B	base	0.779	**0.757**	0.856	0.291	0.553	0.245	0.573	0.345
	instruct	**0.717**	0.785	**0.816**	**0.364**	**0.474**	**0.307**	**0.735**	**0.439**
Qwen2.5-32B	base	0.726	**0.726**	0.830	0.351	0.473	0.282	0.721	0.382
	instruct	**0.674**	0.759	**0.787**	**0.418**	**0.437**	**0.333**	**0.840**	**0.472**
OLMo 3 7B	base	**0.794**	**0.761**	**0.850**	**0.267**	**0.568**	**0.225**	0.470	0.256
	Instruct (RLVR)	0.823	0.851	0.884	0.257	0.604	0.220	**0.485**	**0.317**
OLMo 3.1 32B	base	0.746	**0.743**	0.835	0.334	0.497	0.273	0.652	0.370
	Instruct (RLVR)	**0.701**	0.763	**0.812**	**0.391**	**0.467**	**0.312**	**0.779**	**0.392**
Chance	patient-blind	0.794	0.774	—	0.238	—	0.272	0.102	0.149
Oracle	reference	0.265	0.18–0.25	—	0.735	0.000	—	—	0.975
	one answer: reference mode	—	0.694	—	—	—	—	—	—
	one answer: true pathology	—	0.714	—	—	—	—	—	—

Table 2: Stated vs. sampled readouts on DDXPlus (T=0.7, 980 cases, valid samples). *One list*: one sample's stated differential (per-case mean over valid samples). *Samples*: empirical distribution of top answers. *List w/o gold*: TV after removing the true pathology from list and reference. *Cond. TV*: TV between the list and the reference restricted to the named conditions and renormalized (how the list allocates probability, independent of how much mass it names). DDF1: set F1 of the named conditions against the reference's support. *Recall@5*: the list contains the true pathology. *Severe named*: share of severe conditions with reference probability ≥ 0.1 that one list names. Chance: the top five of the average reference differential as a fixed list, and 32 draws from that average as samples. Oracle: the reference's own top five as the list and as many reference draws as valid samples; and samplers that always give one answer, each case's reference mode or its true pathology. Bold: better model of each pair.

Which signal to use. Across 56 model–setting combinations evaluated on held-out cases (Appendix G.3), combining helps post-trained models only modestly: against agreement across 31 samples (AUROC 0.742 for the 39 post-trained combinations), averaging the two signals' ranks without labels adds 0.019 $[-0.007, 0.040]$ and a logistic combination fitted on 100 labeled cases 0.030 $[0.013, 0.044]$, while the stated probability alone gives 0.671; for base models agreement alone is as good (intervals resample the 26 underlying models).

4.4. Real rare-disease cases and medical post-training

Real rare-disease cases. Phenopacket Store has no reference differential, and 7–8B models are near the floor (pass@32 2.5–7.0%, against about 20% top-1 for GPT-4o (Chimirri et al., 2025)), so we also count closely related Mondo classes as correct (Appendix H.3) and read these results as directional. The voting and agreement shifts replicate: majority voting helps the base models by 3.1–4.6 points but the final post-trained models by only 0.4–1.5, and post-training weakens agreement (AUROC 0.81–0.85 → 0.73–0.76) and strengthens the stated probability (0.46–0.60 → 0.65–0.72). Unlike on DDXPlus, agreement stays the better signal and the three most frequent sampled answers still beat one list's top three (Qwen3-8B: a tie).

Medical models. Medical post-training on Qwen2.5-7B-Instruct does not narrow the samples further: AlphaMed improves on its parent at large k (+8.4 [2.6, 14.7] points at k=32) with unchanged diversity, and HuatuoGPT-o1 and m1, trained on reasoning traces, widen the samples (2.49 and 3.29 distinct answers, against 1.82; Appendix H.1). On Phenopacket Store the RL-trained medical models raise stated confidence from 0.51 to 0.67–0.69 while accuracy gains stay under 1.5 points (Appendix H.2).

5. Discussion and limitations

Jang et al. (2026) show that instruction-tuned models cannot sample from distributions they can describe. In differential diagnosis, against a reference with a known true pathology, we find the same split and a simpler account of it: direct-answer post-training keeps the base model's view of each patient and removes most of its variation between calls. The sampled answers therefore collapse, and evaluations that read the differential from samples (pass@k at large k, diversity, self-consistency, agreement-based uncertainty) report a loss that a clinician reading one answer would not see; combining agreement with the stated probability is a modest improvement for post-trained models, and agreement remains the better signal for base models and reasoning modes. For Qwen, one post-trained call names the true pathology and the reference's conditions more often than one sampled base call, but no more often than a few averaged base calls, and its first-answer advantage largely disappears against a greedily decoded base model with a matched prompt, as the sharpening account predicts (Huang et al., 2024), so it is no evidence that post-training adds diagnostic knowledge. Asking for must-not-miss conditions raises severe coverage, partly through longer lists. Every model stays far from the reference, neither readout tracks how ambiguous a case is (Table 25), and losses of specific severe conditions, notably acute pulmonary edema, need checking model by model. Thinking mode restores pass@32 and agreement, though not breadth, and hotter sampling does so for Qwen2.5-7B-Instruct.

Limitations. DDXPlus is synthetic and closed-set: its reference differential is a rule-based engine's posterior over questionnaire answers without exam or test results, sometimes clinically implausible (Appendix B), its test set is class-balanced, and the prompt lists all 49 conditions, so our metrics measure agreement with that engine, not clinical correctness, and no clinician reviewed lists or references. Severity is the dataset's label or a published list mapped onto its 49 conditions, and the prompt asks for the most likely conditions, not must-not-miss ones. The few-shot controls use one set of example patients, and the temperature controls cover the 7–8B models only. We study open 7–32B models from three families, one reasoning mode and no frontier models. Phenopacket accuracy is near the floor and its grader is not validated against clinicians.

Statement on AI Use. An AI assistant (Claude, Anthropic) was used to write and run analysis code, to draft and edit text, and to check citations. The authors designed the study, reviewed all code, results and text, and take responsibility for the content.

References

Shivam Agarwal, Zimin Zhang, Lifan Yuan, Jiawei Han, and Hao Peng. The unreasonable effectiveness of entropy minimization in LLM reasoning, 2025.

Kasra Arabi, Virginia Smith, and Chhavi Yadav. Sample what you say: Aligning language models to sample the distributions they state, 2026.

Junying Chen, Zhenyang Cai, Ke Ji, Xidong Wang, Wanlong Liu, Rongsheng Wang, and Benyou Wang. Towards medical complex reasoning with LLMs through medical verifiable problems. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 14552–14573, Vienna, Austria, 2025. Association for Computational Linguistics. doi: 10.18653/v1/2025.findings-acl.751. arXiv version titled "HuatuoGPT-o1, Towards Medical Complex Reasoning with LLMs".

Mark Chen, Jerry Tworek, Heewoo Jun, Qiming Yuan, Henrique Ponde de Oliveira Pinto, Jared Kaplan, Harri Edwards, Yuri Burda, Nicholas Joseph, Greg Brockman, Alex Ray, Raul Puri, Gretchen Krueger, Michael Petrov, Heidy Khlaaf, Girish Sastry, Pamela Mishkin, Brooke Chan, Scott Gray, Nick Ryder, Mikhail Pavlov, Alethea Power, Lukasz Kaiser, Mohammad Bavarian, Clemens Winter, Philippe Tillet, Felipe Petroski Such, Dave Cummings, Matthias Plappert, Fotios Chantzis, Elizabeth Barnes, Ariel Herbert-Voss, William Hebgen Guss, Alex Nichol, Alex Paino, Nikolas Tezak, Jie Tang, Igor Babuschkin, Suchir Balaji, Shantanu Jain, William Saunders, Christopher Hesse, Andrew N. Carr, Jan Leike, Josh Achiam, Vedant Misra, Evan Morikawa, Alec Radford, Matthew Knight, Miles Brundage, Mira Murati, Katie Mayer, Peter Welinder, Bob McGrew, Dario Amodei, Sam McCandlish, Ilya Sutskever, and Wojciech Zaremba. Evaluating large language models trained on code. 2021.

Zhoujun Cheng, Shibo Hao, Tianyang Liu, Fan Zhou, Yutao Xie, Feng Yao, Yuexin Bian, Yonghao Zhuang, Nilabjo Dey, Yuheng Zha, Yi Gu, Kun Zhou, Yuqi Wang, Yuan Li, Richard Fan, Jianshu She, Chengqian Gao, Abulhair Saparov, Haonan Li, Taylor W. Killian, Mikhail Yurochkin, Zhengzhong Liu, Eric P. Xing, and Zhiting Hu. Revisiting reinforcement learning for LLM reasoning from a cross-domain perspective. In *Advances in Neural Information Processing Systems (NeurIPS 2025), Datasets and Benchmarks Track*, 2025.

Leonardo Chimirri, J. Harry Caufield, Yasemin Bridges, Nicolas Matentzoglu, Michael Gargano, et al. Consistent performance of large language models in rare disease diagnosis across ten languages and 4917 cases. *eBioMedicine*, 121:105957, 2025. doi: 10.1016/j.ebiom.2025.105957.

Mehul Damani, Isha Puri, Stewart Slocum, Idan Shenfeld, Leshem Choshen, Yoon Kim, and Jacob Andreas. Beyond binary rewards: Training LMs to reason about their uncertainty. In *International Conference on Learning Representations (ICLR)*, 2026.

Daniel Danis, Michael J. Bamshad, Yasemin Bridges, Andrés Caballero-Oteyza, Pilar Cacheiro, Leigh C. Carmody, Leonardo Chimirri, Jessica X. Chong, Ben Coleman, Raymond Dalgleish, Peter J. Freeman, Adam S. L. Graefe, Tudor Groza, Peter Hansen,

Julius O. B. Jacobsen, Adam Klocperk, Maaike Kusters, Markus S. Ladewig, Allison J. Marcello, Teresa Mattina, Christopher J. Mungall, Monica C. Munoz-Torres, Justin T. Reese, Filip Rehburg, Bárbara C. S. Reis, Catharina Schuetz, Damian Smedley, Timmy Strauss, Jagadish Chandrabose Sundaramurthi, Sylvia Thun, Kyran Wissink, John F. Wagstaff, David Zocche, Melissa A. Haendel, and Peter N. Robinson. A corpus of GA4GH phenopackets: Case-level phenotyping for genomic diagnostics and discovery. *Human Genetics and Genomics Advances*, 6(1):100371, 2025. doi: 10.1016/j.xhgg.2024.100371.

Marius Dragoi, Ioana Pintilie, Florin Gogianu, and Florin Brad. Beyond pass@k: Breadth-depth metrics for reasoning boundaries, 2025.

Arsene Fansi Tchango, Rishab Goel, Julien Martel, Zhi Wen, Gaetan Marceau Caron, and Joumana Ghosn. Towards trustworthy automatic diagnosis systems by emulating doctors' reasoning with deep reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS 2022)*, volume 35, 2022a.

Arsene Fansi Tchango, Rishab Goel, Zhi Wen, Julien Martel, and Joumana Ghosn. DDXPlus: A new dataset for automatic medical diagnosis. In *Advances in Neural Information Processing Systems (NeurIPS 2022), Datasets and Benchmarks Track*, 2022b.

Sebastian Farquhar, Jannik Kossen, Lorenz Kuhn, and Yarin Gal. Detecting hallucinations in large language models using semantic entropy. *Nature*, 630(8017):625–630, 2024.

Xiangming Gu, Soham De, Michalis Titsias, Larisa Markeeva, Petar Veličković, and Razvan Pascanu. The illusion of stochasticity in LLMs. In *Conference on Language Modeling (COLM)*, 2026. arXiv:2604.06543.

Martha Gulati, Phillip D. Levy, Debabrata Mukherjee, Ezra Amsterdam, Deepak L. Bhatt, et al. 2021 AHA/ACC/ASE/CHEST/SAEM/SCCT/SCMR guideline for the evaluation and diagnosis of chest pain. *Circulation*, 144(22):e368–e454, 2021. doi: 10.1161/CIR.0000000000001029.

Kimia Hamidieh, Veronika Thost, Walter Gerych, Mikhail Yurochkin, and Marzyeh Ghassemi. Complementing self-consistency with cross-model disagreement for uncertainty quantification, 2026.

Audrey Huang, Adam Block, Dylan J. Foster, Dhruv Rohatgi, Cyril Zhang, Max Simchowitz, Jordan T. Ash, and Akshay Krishnamurthy. Self-improvement in language models: The sharpening mechanism, 2024.

Xiaoke Huang, Juncheng Wu, Hui Liu, Xianfeng Tang, and Yuyin Zhou. m1: Unleash the potential of test-time scaling for medical reasoning with large language models. In *Proceedings of the Fifth Machine Learning for Health Symposium (ML4H 2025)*, volume 297 of *Proceedings of Machine Learning Research*, pages 369–383. PMLR, 2026.

Chaemin Jang, Dongman Lee, and Jihee Kim. Instruction-tuned language models cannot sample from distributions they can describe. In *Findings of the Association for Computational Linguistics: EMNLP 2026*, 2026.

Liwei Jiang, Yuanjun Chai, Margaret Li, Mickel Liu, Raymond Fok, Nouha Dziri, Yulia Tsvetkov, Maarten Sap, Alon Albalak, and Yejin Choi. Artificial hivemind: The open-ended homogeneity of language models (and beyond). In *Advances in Neural Information Processing Systems (NeurIPS 2025), Datasets and Benchmarks Track*, 2025.

Zahir Kanjee, Byron Crowe, and Adam Rodman. Accuracy of a generative artificial intelligence model in a complex diagnostic challenge. *JAMA*, 330(1):78, 2023. doi: 10.1001/jama.2023.8288.

Aayush Karan and Yilun Du. Reasoning with sampling: Your base model is smarter than you think, 2025.

Constantinos Karouzos, Xingwei Tan, and Nikolaos Aletras. Where does output diversity collapse in post-training?, 2026.

Robert Kirk, Ishita Mediratta, Christoforos Nalmpantis, Jelena Luketina, Eric Hambro, Edward Grefenstette, and Roberta Raileanu. Understanding the effects of RLHF on LLM generalisation and diversity. In *International Conference on Learning Representations (ICLR)*, 2024.

Thom Lake, Eunsol Choi, and Greg Durrett. From distributional to overton pluralism: Investigating large language model alignment. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 6794–6814, 2025.

Jixuan Leng, Chengsong Huang, Banghua Zhu, and Jiaxin Huang. Taming overconfidence in LLMs: Reward calibration in RLHF. In *International Conference on Learning Representations (ICLR)*, 2025.

Seungseop Lim, Gibaeg Kim, Hyunkyung Lee, Wooseok Han, Jean Seo, Jaehyo Yoo, and Eunho Yang. H-DDx: A hierarchical evaluation framework for differential diagnosis. 2025. NeurIPS 2025 Workshop on Generative AI for Health (GenAI4Health).

Bill Yuchen Lin, Abhilasha Ravichander, Ximing Lu, Nouha Dziri, Melanie Sclar, Khyathi Chandu, Chandra Bhagavatula, and Yejin Choi. The unlocking spell on base LLMs: Rethinking alignment via in-context learning, 2023.

Che Liu, Haozhe Wang, Jiazhen Pan, Zhongwei Wan, Yong Dai, Fangzhen Lin, Wenjia Bai, Daniel Rueckert, and Rossella Arcucci. Beyond distillation: Pushing the limits of medical LLM reasoning with minimalist rule-based RL. 2025. Introduces the AlphaMed model; NeurIPS 2025 Workshop on Generative AI for Health (GenAI4Health).

Mingyi Liu. The alignment tax: Response homogenization in aligned LLMs and its implications for uncertainty estimation, 2026.

Beier Luo, Shuoyuan Wang, Sharon Li, and Hongxin Wei. Your pre-trained LLM is secretly an unsupervised confidence calibrator. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. arXiv:2505.16690.

Qing Lyu, Kumar Shridhar, Chaitanya Malaviya, Li Zhang, Yanai Elazar, Niket Tandon, Marianna Apidianaki, Mrinmaya Sachan, and Chris Callison-Burch. Calibrating large language models with sample consistency, 2024.

Daniel McDuff, Mike Schaekermann, Tao Tu, Anil Palepu, Amy Wang, Jake Garrison, Karan Singhal, Yash Sharma, Shekoofeh Azizi, Kavita Kulkarni, et al. Towards accurate differential diagnosis with large language models. *Nature*, 642(8067):451–457, 2025. doi: 10.1038/s41586-025-08869-4.

Zhiting Mei, Christina Zhang, Tenny Yin, Justin Lidard, Ola Shorinwa, and Anirudha Majumdar. Reasoning about uncertainty: Do reasoning models know when they don't know? In *Findings of the Association for Computational Linguistics: EACL 2026*, pages 3408–3458, Rabat, Morocco, 2026. Association for Computational Linguistics. doi: 10.18653/v1/2026.findings-eacl.178.

Nicole Meister, Carlos Guestrin, and Tatsunori Hashimoto. Benchmarking distributional alignment of large language models. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 24–49, Albuquerque, New Mexico, 2025. Association for Computational Linguistics. doi: 10.18653/v1/2025.naacl-long.2.

Sonia Krishna Murthy, Tomer Ullman, and Jennifer Hu. One fish, two fish, but not the whole sea: Alignment reduces language models' conceptual diversity. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 11241–11258, 2025. doi: 10.18653/v1/2025.naacl-long.561.

David E. Newman-Toker, Najlla Nassery, Adam C. Schaffer, Chihwen Winnie Yu-Moe, Gwendolyn D. Clemens, et al. Burden of serious harms from diagnostic error in the USA. *BMJ Quality & Safety*, 33(2):109–120, 2024. doi: 10.1136/bmjqs-2021-014130.

Changdae Oh, Qi Zeng, Qi Qi, Andrey Zhmoginov, Deren Lei, Yun He, Hoang Phan, Hangoo Kang, Azalia Mirhoseini, and Sharon Li. Sharpening tax in post-training. 2026.

OpenAI. GPT-4 technical report, 2023.

Max Peeperkorn, Tom Kouwenhoven, Dan Brown, and Anna Jordanous. Mind the gap: Conformative decoding to improve output diversity of instruction-tuned large language models, 2025.

Archiki Prasad, Weizhe Yuan, Richard Yuanzhe Pang, Jing Xu, Maryam Fazel-Zarandi, Mohit Bansal, Sainbayar Sukhbaatar, Jason Weston, and Jane Yu. Self-consistency preference optimization. In *International Conference on Machine Learning (ICML)*, 2025. arXiv:2411.04109.

Isha Puri, Mehul Damani, Idan Shenfeld, Marzyeh Ghassemi, Jacob Andreas, and Yoon Kim. Escaping the mode: Multi-answer reinforcement learning in LMs. In *Proceedings of the 43rd International Conference on Machine Learning (ICML)*, volume 306 of *Proceedings of Machine Learning Research*, pages 100127–100146. PMLR, 2026. arXiv

version titled "Reaching Beyond the Mode: RL for Distributional Reasoning in Language Models".

Justin T. Reese, Leonardo Chimirri, Yasemin Bridges, Daniel Danis, J. Harry Caufield, Michael A. Gargano, Carlo Kroll, Andrew Schmeder, Fengchen Liu, Kyran Wissink, Julie A. McMurry, Peter N. Robinson, et al. Systematic benchmarking demonstrates large language models have not reached the diagnostic accuracy of traditional rare-disease decision support tools. *European Journal of Human Genetics*, 34(4):498–504, 2026. doi: 10.1038/s41431-026-02054-5.

Zhiyao Ren, Yibing Zhan, Siyuan Liang, Guozheng Ma, Baosheng Yu, and Dacheng Tao. Towards reliable medical LLMs: Benchmarking and enhancing confidence estimation of large language models in medical consultation, 2026.

Daniel Philip Rose, Chia-Chien Hung, Marco Lepri, Israa Alqassem, Kiril Gashteovski, and Carolin Lawrence. MEDDxAgent: A unified modular agent framework for explainable automatic differential diagnosis. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 13803–13826, 2025. doi: 10.18653/v1/2025.acl-long.677.

Thomas Savage, John Wang, Robert Gallo, Abdessalem Boukil, Vishwesh Patel, Seyed Amir Ahmad Safavi-Naini, Ali Soroush, and Jonathan H. Chen. Large language model uncertainty proxies: discrimination and calibration for medical diagnosis and treatment. *Journal of the American Medical Informatics Association*, 32(1):139–149, 2025. doi: 10.1093/jamia/ocae254.

Yuda Song, Julia Kempe, and Remi Munos. Outcome-based exploration for LLM reasoning, 2025.

Jacob Mitchell Springer, Madhu Advani, Lukas Aichberger, Arwen Bradley, Eran Malach, Omid Saremi, Sinead Williamson, Preetum Nakkiran, Etai Littwin, and Aditi Raghunathan. Annotations mitigate post-training mode collapse. In *Proceedings of the 43rd International Conference on Machine Learning (ICML)*, 2026.

Hexiang Tan, Fei Sun, Sha Liu, Du Su, Qi Cao, Xin Chen, Jingang Wang, Xunliang Cai, Yuanzhuo Wang, Huawei Shen, and Xueqi Cheng. Too consistent to detect: A study of self-consistent errors in LLMs. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2025. arXiv:2505.17656.

Amir Taubenfeld, Tom Sheffer, Eran Ofek, Amir Feder, Ariel Goldstein, Zorik Gekhman, and Gal Yona. Confidence improves self-consistency in LLMs. In *Findings of the Association for Computational Linguistics: ACL 2025*, 2025. arXiv:2502.06233.

Team Olmo, Allyson Ettinger, Amanda Bertsch, Bailey Kuehl, David Graham, David Heineman, Dirk Groeneveld, Faeze Brahman, Finbarr Timbers, Hamish Ivison, et al. Olmo 3, 2025.

Katherine Tian, Eric Mitchell, Allan Zhou, Archit Sharma, Rafael Rafailov, Huaxiu Yao, Chelsea Finn, and Christopher D. Manning. Just ask for calibration: Strategies for

eliciting calibrated confidence scores from language models fine-tuned with human feedback. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023.

Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. In *International Conference on Learning Representations (ICLR)*, 2023.

Peter West and Christopher Potts. Base models beat aligned models at randomness and creativity. In *Conference on Language Modeling (COLM)*, 2025.

Fang Wu, Weihao Xuan, Danny Xing, Ximing Lu, Mingjie Liu, Yi Dong, Zaid Harchaoui, and Yejin Choi. The invisible leash: Why RLVR may or may not escape its origin. 2025.

Miao Xiong, Zhiyuan Hu, Xinyang Lu, Yifei Li, Jie Fu, Junxian He, and Bryan Hooi. Can LLMs express their uncertainty? an empirical evaluation of confidence elicitation in LLMs. In *International Conference on Learning Representations (ICLR)*, 2024.

Chenghao Yang, Sida Li, and Ari Holtzman. LLM probability concentration: How alignment shrinks the generative horizon. *Transactions on Machine Learning Research*, 2026. arXiv:2506.17871.

Yang Yue, Zhiqi Chen, Rui Lu, Andrew Zhao, Zhaokai Wang, Yang Yue, Shiji Song, and Gao Huang. Does reinforcement learning really incentivize reasoning capacity in LLMs beyond the base model? In *Advances in Neural Information Processing Systems (NeurIPS 2025)*, 2025.

Longfei Yun, Chenyang An, Zilong Wang, Letian Peng, and Jingbo Shang. The price of format: Diversity collapse in LLMs, 2025.

Jiayi Zhang, Simon Yu, Derek Chong, Anthony Sicilia, Michael R. Tomz, Christopher D. Manning, and Weiyan Shi. Verbalized sampling: How to mitigate mode collapse and unlock LLM diversity. In *Proceedings of the 43rd International Conference on Machine Learning (ICML)*, volume 306 of *Proceedings of Machine Learning Research*, pages 160073–160152. PMLR, 2026.

Minda Zhao, Yilun Du, and Mengyu Wang. Large language models are bad dice players: LLMs struggle to generate random numbers from statistical distributions. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2026. arXiv:2601.05414.

Chiwei Zhu, Benfeng Xu, Quan Wang, Yongdong Zhang, and Zhendong Mao. On the calibration of large language models and alignment. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, 2023. arXiv:2311.13240.

Appendix A. Claims and the evidence for them

Table 3 lists each claim with its primary evidence, the checks under which it holds and those under which it weakens or fails. Primary evidence comes from the 112 headline tests (§3); everything else is a secondary check, and Table 13 gives the same information control by control.

Appendix B. Prompts, vignettes and grading

DDXPlus. Each patient's coded evidence is rendered by a fixed template into a vignette of the patient's answers and relevant history; we fix known mistranslations in the English evidence descriptions. An example (test case with true pathology chronic rhinosinusitis; the reference differential gives viral pharyngitis 0.17, chronic rhinosinusitis 0.16, cluster headache 0.15, acute rhinosinusitis 0.14, bronchitis 0.14, URTI 0.12; cluster headache is implausible for this patient, a reminder that the reference is an engine's posterior rather than a clinician's differential):

> `A 13-year-old female presents for evaluation. Symptoms (patient answers): Do you have pain somewhere, related to your reason for consulting? Yes. Characterize your pain: burning. Do you feel pain somewhere? mouth, forehead, cheek(R), cheek(L), nose. How intense is the pain? (scale 0-10) 3. […] Have you lost your sense of smell? Yes. Do you have nasal congestion or a clear runny nose? Yes. Do you have a cough? Yes. Relevant history and risk factors: Have you had a cold in the last 2 weeks? Yes. […] Do you have polyps in your nose? Yes. […]`

System prompt: `You are an experienced emergency and primary care physician.` The user turn is the vignette followed by: `Based on the patient's answers above, give your differential diagnosis. Choose ONLY from this list of possible conditions: [the 49 conditions, one per line]. List the 5 most likely conditions from the list, most likely first, with your probability that each is the correct diagnosis. Use the condition names exactly as written in the list. End your answer with a JSON array in exactly this format: [{"diagnosis": "<name>", "probability": <number between 0 and 1>}, ...].` The full-distribution prompt (§E.1) instead asks for a probability on every condition, summing to one, most likely first. The plain and CoT variants are described in §D.3. Base models receive the same chat-formatted prompt; OLMo base models use their SFT checkpoint's chat template.

Phenopacket Store. The vignette lists the case's present and (up to ten) excluded HPO phenotypes with sex and age. System prompt: `You are an expert clinical geneticist and diagnostician.` Instructions: `List the 5 most likely specific diagnoses (use the specific disease or subtype name, e.g. 'Spastic paraplegia 7' rather than 'hereditary spastic paraplegia', when you can), most likely first, with your probability that each is the correct diagnosis`, followed by the same JSON format.

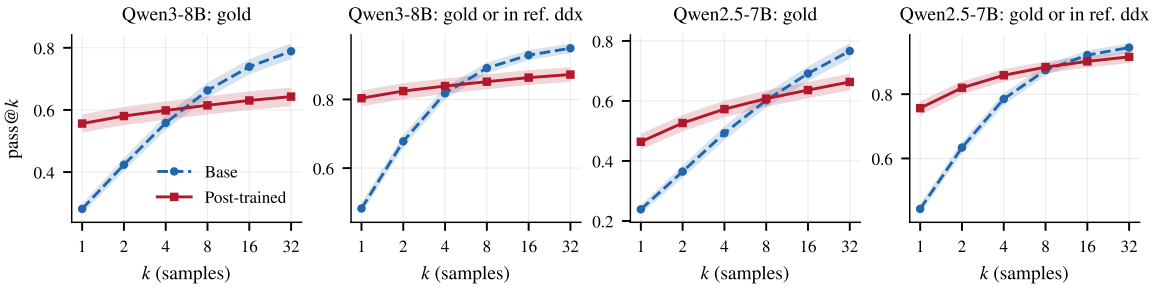


Figure 2: pass@k on DDXPlus at T=1 (980 cases, 32 samples), base model (dashed) vs. its post-trained version (solid), with 95% case-bootstrap bands. "Gold": the top-1 answer is the true pathology; "gold or in ref. ddx": it is the true pathology or appears in the reference differential. In both families the post-trained model wins at small k and loses at large k.

Grading open-ended names. Each predicted name is mapped to Mondo by exact lookup against Mondo labels and synonyms (also trying any parenthesized abbreviation and the name without it), then by fuzzy matching (token-sort ratio ≥ 92). It is *exact* if it maps to the Mondo class of the case's OMIM diagnosis, and *related* if it maps to a specific ancestor group of it (at most two is-a steps up, with at most 150 descendants) or to one of its subtypes; siblings sharing only a broad parent are wrong. The grader passes hand-written checks (e.g. "Hereditary spastic paraplegia type 7 (SPG7)" is exact for SPG7, "Hereditary spastic paraplegia" is related, and Turner syndrome is wrong for Jacobsen syndrome), but we did not validate it against clinician grading, so errors at the scale of the 1–2 point effects we report on this dataset are possible.

Appendix C. Models

Appendix D. The sampled differential in detail

D.1. General post-training trades coverage for top-1 accuracy

Table 5 and Figure 2 compare each base model with its post-trained counterpart on DDXPlus. In both families the post-trained model is much better on its first answer: +27.5 points for Qwen3 and +22.5 for Qwen2.5 at T=1 (intervals $[+25.3, +29.6]$ and $[+20.7, +24.4]$). The advantage shrinks with k and reverses: the curves cross at k≈4–8, and at k=32 the base model leads by 14.6 points (Qwen3, $[+12.2, +17.1]$) and 10.3 points (Qwen2.5, $[+7.8, +12.8]$). The pattern is the same at T=0.7 for Qwen3, crossing at k=4 with a base lead of 12.4 points at k=32 ($[+10.1, +14.9]$). Unparsable base-model samples count as wrong, so these gaps are lower bounds. Qwen2.5 at T=0.7 behaves the same way: +12.6 points at k=1 $[+10.7, +14.5]$ and a base lead of 9.7 at k=32 $[+7.3, +12.0]$.

D.2. The sampled differential collapses, while top-1 calibration improves

The gap at large k reflects a collapsed *sampled* differential. Across 32 samples, the post-trained Qwen models name 1.3–2.7 distinct diagnoses per case, against 3.8–8.2 for their base models. Their answer entropy is 0.11–0.49, against 0.78–1.58 for base models and 1.87 for the reference differential (Table 5). Sampling at T=1 widens the base models' answers but barely changes the post-trained models' (Qwen3 entropy 0.11 at T=0.7, 0.18 at T=1), so the collapse lives in the model's own distribution rather than in the decoding settings. As a result, post-trained models cover less of the reference differential: 0.24 vs. 0.41 of its probability mass at k=32 for Qwen3, and 0.29 vs. 0.42 for Qwen2.5. Their top-1 calibration, however, is markedly better (ECE 0.06 vs. 0.21 and 0.14 vs. 0.30). This concerns verbalized confidence; alignment is known to worsen the calibration of token probabilities (OpenAI, 2023; Zhu et al., 2023), while verbalized confidence of RLHF models is better calibrated (Tian et al., 2023). On every single-answer criterion (accuracy, small-k pass@k, calibration), the post-trained model therefore looks strictly better, and the loss of breadth in its samples is invisible. Whether the breadth is lost from the model's stated differential as well is a separate question (§4.2).

Which diagnoses drop out? For each case and each condition that the reference differential weights at least 0.1, we ask whether the model names that condition as its top answer in at least one of its 32 samples. We call this *considering* the condition. We group the 49 conditions by DDXPlus severity (1 = most severe). Post-training lowers consideration in every tier (Table 6), and the severe tier (DDXPlus severity 1–2, the dataset's label rather than a clinical consensus list) is not spared. This metric needs a chance reference, though: a guesser that picks uniformly among the 49 conditions scores about 0.48 on it, and one that draws from the average reference differential 0.56–0.61, above every model, base models included. Considered@32 therefore measures the breadth of the sampled answers, not clinical judgement; we assess severe conditions on the stated list instead (§E.2). For the 17 severity-1–2 conditions (e.g. NSTEMI/STEMI, acute pulmonary edema, anaphylaxis, pulmonary embolism, unstable angina), consideration falls from 0.45 to 0.27 for Qwen3 and from 0.51 to 0.34 for Qwen2.5 at T=1. The differences are -18.4 $[-20.8, -15.9]$ and -17.0 $[-19.7, -14.2]$ points. Some diagnoses disappear from the top answer almost entirely. Qwen2.5-7B-Instruct names unstable angina as its top answer in one of 107 eligible cases, where its base model does in 47%. When unstable angina is the true pathology, pass@32 falls from 1.00 to 0.00. Stable angina (0.89 to 0.06) and myocarditis (0.41 to 0.00) follow the same pattern. These conditions are not forgotten: they still appear lower in the listed differential (unstable angina in 50% of eligible cases, against 95% for the base model). The model ranks them but no longer commits to them. At the same time, post-trained models concentrate on a few salient "attractor" diagnoses. Qwen2.5-7B-Instruct gives NSTEMI/STEMI as its top answer in 13.1% of samples, against a reference rate of 4.4% and 6.4% for its base model. It also over-predicts pulmonary embolism (8.2% vs. 3.2%) and pneumonia (8.0% vs. 2.4%). Chest-pain presentations thus collapse onto the most dangerous candidate. That raises its accuracy (NSTEMI pass@1 0.99 vs. 0.44) at the cost of the alternatives. Ruling out acute coronary syndrome first is standard practice for chest pain, but these vignettes lack the ECG and biomarkers that would separate the coronary conditions, so we read this as a concentration of the samples, not as a clinical error.

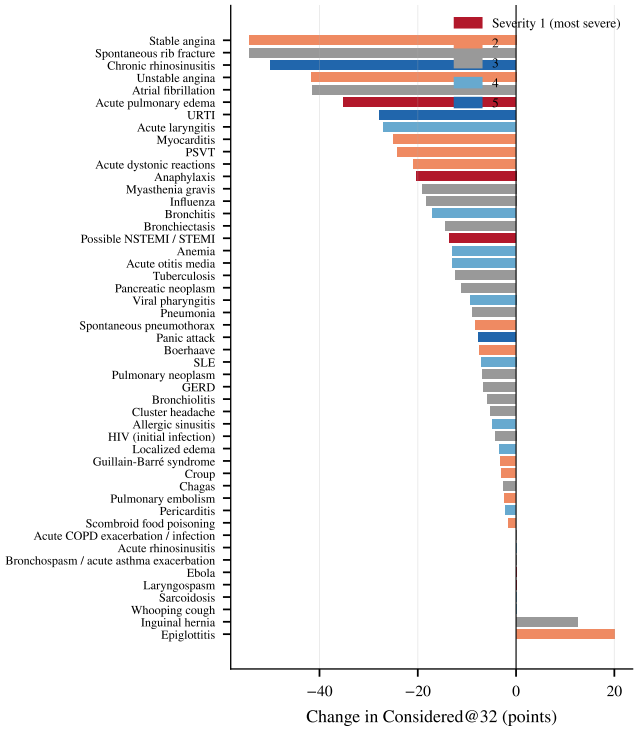


Figure 3: Change in Considered@32 per condition, Qwen2.5-7B-Instruct minus its base model (T=1, 980 cases), colored by DDXPlus severity (dark red = most severe). Conditions with fewer than five cases giving them reference probability ≥ 0.1 are omitted. Angina, acute pulmonary edema, anaphylaxis and myocarditis are among the largest drops.

D.3. Robustness to prompt and output format

Prompt. We repeat the Qwen3-8B pair with two other prompts (Table 7). *Plain* drops the system prompt and rewords the instructions. *CoT* asks the model to reason step by step before answering; Qwen3's thinking mode stays off in all runs. The plain prompt reproduces the pattern: the post-trained model gains +23.0 [+20.9, +25.1] points at k=1 and loses −7.1 [−9.3, −5.0] at k=32 (T=0.7; at T=1, +34.2 and −8.7 [−11.0, −6.4]), with 1.3–1.5 distinct answers against 3.7–5.6. Asking for reasoning changes the picture. With CoT, the post-trained model's pass@32 rises from 0.607 to 0.835: it leads the base model at every k at T=0.7 (+4.7 [+2.9, +6.6] at k=32) and ties it at T=1 (0.0 [−2.1, +1.9]). Its samples remain far less diverse than the base model's (2.1–2.2 vs. 3.9–6.3 distinct answers), so reasoning recovers coverage at k=32 without restoring sampling diversity. The coverage loss is therefore largest when the model answers directly, as in most clinical decision-support deployments, and reasoning at inference time mitigates it. This matches the medical models: the reasoning-trained descendants widen the differential (§H.1).

Output format. Base models more often give invalid answers: unparsable, or with a top answer outside the 49 conditions. This affects 12% of Qwen3-8B-Base samples at T=0.7 and 25% at T=1 (Qwen2.5-7B: 1% and 21%), against at most 0.3% for the post-trained models. Invalid samples count as wrong, which deflates the base model at large k but inflates the post-trained model's gain at small k. Among valid samples only, the post-trained models still gain +12.0 [+9.8, +14.1] (Qwen3, T=0.7), +18.7 [+16.6, +20.7] (Qwen3, T=1), +12.2 [+10.4, +14.0] and +16.8 [+15.0, +18.5] points (Qwen2.5) at k=1, so a quarter to a third of the raw gain at T=1 is format. On the cases where every sample of both models is valid, the large-k loss persists: −15.2 [−21.9, −8.6] at k=32 for Qwen3 (105 cases) and −10.2 [−12.7, −7.6] for Qwen2.5 (748 cases), both at T=0.7. (At T=1 almost no case has 32 valid base samples.) The reported large-k losses are thus lower bounds.

D.4. Which post-training stage narrows the differential?

The Qwen pairs compare a base model with the end of a post-training pipeline. OLMo 3 (Team Olmo et al., 2025) releases every stage of its 7B instruct pipeline from one base model: supervised fine-tuning, then DPO, then RL with verifiable rewards on math, code, instruction-following and chat prompts (the released OLMo-3-7B-Instruct). It also releases RL-Zero checkpoints trained with RL directly from the base; we use the general one, trained on chat prompts with an LM-judge reward. (The RL-Zero-Mix checkpoint does not load in our vLLM version.) Table 8 gives the results.

The coverage loss replicates in OLMo. At T=0.7, every post-trained stage is worse than the base at k=32: by −14.6 [−17.5, −11.7] points after SFT, −9.2 [−12.1, −6.1] after DPO, −28.3 [−31.5, −25.0] after RLVR, and −13.3 [−16.1, −10.5] for RL-Zero. Diversity falls at every stage of the pipeline, from 4.93 distinct answers for the base to 4.01 after SFT, 3.30 after DPO and 2.37 after RLVR.

Compared stage to stage, DPO recovers part of what SFT lost at large k (+5.4 [+2.9, +8.0] points at k=32, and +10.3 [+9.0, +11.6] at k=1) while narrowing the differential further (−0.71 distinct answers). The final RLVR stage is the largest single narrowing. Against the DPO model it loses −19.1 [−21.8, −16.3] points at k=32, 0.92 distinct answers and 7.7 points of coverage@32, and it is also worse at k=1 (−5.9 [−7.2, −4.6]). It drops severe-tier Considered@32 to 0.25, a loss of −16.1 [−18.6, −13.5] points relative to the base, against −7.4 for the DPO model. At T=1 the pattern is the same (RLVR vs. DPO: −12.7 [−15.4, −9.9] at k=32).

RL applied directly to the base narrows the differential less. At T=1, RL-Zero shows no detectable loss of the base model's pass@32 (−1.8 [−4.5, +0.8]; clustered [−5.8, +2.1]) and 6.28 of its 7.53 distinct answers, and it loses 4.0 [1.2, 6.9] points of severe-tier consideration, against 14.4 for the full instruct pipeline. At T=0.7 it still loses 13.3 points at k=32 (clustered [7.3, 19.7]), about as much as SFT.

The OLMo base model often breaks the output format: 40% of its samples are invalid at T=0.7 and 47% at T=1, while unparsable completions are 16–21% for the SFT model and at most 1.6% for the later stages. No case has 32 valid base samples, so the all-valid check of §D.3 is not possible here. Among valid samples, the post-trained models' gains at k=1 are small (−1.9 to +4.0 points at T=0.7, +1.3 to +8.1 at T=1), so most of OLMo's raw gain at k=1 is format. Invalid samples count as misses and are excluded from distinct

answers and entropy, so the base model's coverage and diversity are understated and the losses above are conservative.

Across these pipelines, then, the collapse of the sampled differential is not a necessary effect of RL. It accumulates over the instruction-tuning pipeline; in OLMo 3 7B its largest step is RL applied on top of SFT and DPO, but at 32B it is SFT (§D.5), so which stage narrows most depends on the recipe. RL applied directly to a base model (OLMo RL-Zero at T=1, AlphaMed's base-initialized checkpoint, and our GRPO runs in §I) narrows the differential less, but not always: RL-Zero at T=0.7 loses about as much as SFT. On Phenopacket Store the instruct pipeline also loses coverage: every stage trails the base model at k=32 (-2.0 [$-3.0, -1.1$] points after SFT, -1.5 after DPO, -1.4 after RLVR; all significant), while RL-Zero does not (-0.6 [$-1.4, +0.2$]), from a base model at 4.5% (§H.2).

D.5. Replication at 32B

We repeat the DDXPlus comparisons at T=0.7 with Qwen2.5-32B and its Instruct model, and with every released stage of OLMo 3.1 32B, whose base is OLMo-3-1125-32B; there is no 32B RL-Zero checkpoint (Table 9). The sampled differential collapses at 32B too, by about half as much. Post-training costs 4.9 [2.8, 7.1] points at pass@32 for Qwen2.5-32B and 4.0 [2.0, 5.9] for OLMo 3.1 32B (clustered [-0.5, 10.3] and [1.2, 6.8]; neither survives correction), halves the number of distinct answers (3.78 $\rightarrow$ 1.68 and 4.32 $\rightarrow$ 2.15), and lowers severe-tier consideration by 6.1 and 8.4 points. For Qwen the loss holds on the 421 cases where every base sample is valid (-4.8 [$-7.8, -1.7$]); the OLMo 32B base gives invalid answers on 27% of samples. With few-shot prompts, which nearly remove invalid answers, both losses are three to five times larger (§F.1). The stage attribution differs from 7B. In OLMo 3.1 32B the largest narrowing is SFT (pass@32 -12.1 [$-14.4, -9.9$], clustered [$-18.0, -6.8$], distinct answers -1.39, severe tier -11.9); DPO recovers most of it ($+10.9$ against SFT), and the final RL step is small (-2.8 [$-4.3, -1.1$] at k=32, clustered [$-5.3, -0.5$], -0.14 distinct answers). OLMo 3.1 32B and OLMo 3 7B were post-trained with different recipes, so the stage that narrows the samples most is a property of the pipeline, not of SFT, DPO or RL as such. Under this prompt the stated readout improves in both 32B families (§4.2; with few-shot prompts only Qwen2.5-32B's does robustly, §F.1), and voting stops paying off: majority voting over 32 samples gains the base models 7.0 and 8.4 points and the Instruct models 1.9 and 0.9 (the latter n.s.).

Appendix E. The stated readout: elicitation and robustness

E.1. Robustness of the stated readout to elicitation

The stated readout above uses a five-item list. We repeat the Qwen pairs and OLMo 3 7B's base and Instruct models at T=0.7 with a prompt that asks for a probability on every one of the 49 conditions, summing to one, and take each sample's top answer to be its highest-probability condition (Table 10). The results match the short list. After post-training the stated distribution is closer to the reference (TV -0.052 for Qwen3-8B, -0.067 for Qwen2.5-7B-Instruct, -0.032 for OLMo-3-7B-Instruct) while the samples move away ($+0.012$ to $+0.038$). Qwen3-8B's stated probability again beats agreement over 32 samples ($+0.069$ [$+0.033, +0.104$]), Qwen2.5-7B-Instruct's again does not (-0.074), and

OLMo-3-7B-Instruct's ties it (−0.010; +0.044 against five samples). The top three of one full distribution beats the three most frequent of 32 sampled answers for Qwen3-8B (+14.6 points) and narrowly for Qwen2.5-7B-Instruct (+1.6), but not for OLMo-3-7B-Instruct (−9.5).

E.2. Ruling out format, accuracy and floor effects

Table 12 tests simpler explanations of the stated-list result. *Format.* The general models' post-trained lists and base-model lists at T=0.7 are nearly all complete (81–99% name five on-list conditions with probabilities; Qwen base models at T=1, 61–62%; the medical models, 61–80%). Restricting to complete lists leaves the gaps unchanged (Qwen2.5-7B: 0.779 vs. 0.719). *Accuracy.* A more accurate model puts more stated mass on the true pathology, which the reference also favors. Within samples whose top answer is correct, and within those whose top answer is wrong, the post-trained Qwen list is still closer to the reference (−0.018 to −0.034 and −0.036 to −0.079 in TV across both temperatures; Table 12 shows T=0.7). *Floor.* One list can do no better than the reference's own top five, and a finite set of samples no better than as many draws from the reference. Measured as excess over these floors, post-training lowers the list's TV (Qwen3-8B 0.479 → 0.435) and raises the samples' (0.557 → 0.613). *True pathology removed.* With the true pathology removed from both list and reference and both renormalized, the post-trained Qwen and OLMo 3.1 32B lists are still closer to the reference (−0.018 to −0.056), and DDXPlus's set F1 against the reference's support improves by 0.040–0.083 (Table 30). TV, mass and DDF1 are not independent, since a list's TV is the reference mass it leaves out plus the mass it misallocates among the conditions it names, and the second term grows mechanically with the first. The conditional TV, between the list and the reference restricted to the named conditions and renormalized, removes this: it falls after post-training for every Qwen pair and OLMo 3.1 32B (−0.025 to −0.078, all surviving correction) and rises for OLMo 3 7B at T=1 (+0.064), so the Qwen lists both name better conditions and weight them more like the reference. *Patient-blind list.* A fixed list of the five conditions with the highest average reference probability scores TV 0.794, mass 0.238 and DDF1 0.272. Every model's list beats it in TV except OLMo 3 7B's and Qwen2.5-7B's base list at T=1; in DDF1 several base lists do not (Qwen2.5-7B 0.245, OLMo 3 7B 0.225), and the post-trained Qwen and OLMo 3.1 32B lists exceed it by 0.035–0.061. *Over-naming.* As a share of named conditions, post-trained lists name fewer that the reference gives less than 1% (Qwen3-8B 0.59 → 0.55) and fewer unwarranted severe conditions (Qwen2.5-32B 0.74 → 0.65), but as a count per list the change is mixed: from 0.22 fewer unwarranted severe conditions (OLMo 3.1 32B) to 0.35 more (OLMo 3 7B). *Severe conditions one by one.* Table 11 tests each of the 17 severe conditions, counting it wherever the reference gives it any weight. After Benjamini–Hochberg correction, 2–7 conditions per zero-shot pair are named significantly less often after post-training and 0–6 significantly more often, and no pair names more conditions less often than a random swap of base and post-trained lists would. That count test is weak against an overall upward shift, so Table 11 also gives the mean shift across the 17 conditions: it rises significantly in seven of eight zero-shot settings but falls with worked examples for Qwen2.5-32B (−0.012) and OLMo 3 7B (−0.018), and is flat for Mistral (−0.009, p=0.05). Significant losses include acute pulmonary edema (Qwen3-8B 0.75 → 0.61, Qwen2.5-32B

0.63 → 0.44, OLMo 3 7B 0.56 → 0.31, OLMo 3.1 32B 0.57 → 0.41), stable angina (Qwen2.5-7B 0.25 → 0.08), NSTEMI/STEMI (OLMo 3 7B 0.31 → 0.14, OLMo 3.1 32B 0.49 → 0.36) and myocarditis (Qwen2.5-32B 0.14 → 0.05); significant gains include pulmonary embolism (Qwen2.5-7B 0.29 → 0.56, OLMo 3 7B 0.20 → 0.65), NSTEMI/STEMI (Qwen2.5-7B 0.32 → 0.52) and unstable angina (Qwen3-8B 0.14 → 0.37). Among single stages, OLMo 3 7B's RLVR step and OLMo 3.1 32B's SFT and RLVR steps do name more conditions less often than chance (12–13 of 17, p=0.03–0.04).

Sampling from the list. Replacing each sample's top answer with a draw from its own stated list raises pass@32 for every model. Compared like for like, post-trained lists reach higher than their base model's lists in five of seven comparisons (Qwen3-8B 0.811 vs. 0.780 at T=0.7, Qwen2.5-32B-Instruct 0.880 vs. 0.850, OLMo 3.1 32B Instruct 0.853 vs. 0.839), but not for Qwen3-8B at T=1 (0.820 vs. 0.832) or OLMo 3 7B (0.577 vs. 0.719), and a uniform guesser reaches 0.48. OLMo 3 7B remains the exception throughout: its Instruct list is closer to the reference when its top answer is correct but much farther when it is wrong.

Appendix F. Controls: prompt, temperature, decoding, severity labels and a third family

This section collects the controls for the main comparison. Table 13 shows, for each control, whether each main claim holds; the subsections give the numbers. Real-case evidence is in Appendix H.

F.1. Few-shot elicitation

The main analyses give every model the same zero-shot chat prompt, which base models were not trained to follow; they give invalid answers on up to 47% of samples, which we count as wrong. We therefore re-ran the five final base/post-trained pairs at T=0.7 with three worked examples in the prompt: three fixed DDXPlus training patients with different true pathologies, each followed by the top five conditions of its reference differential, with probabilities, in the JSON format the instructions ask for. Base models see the instructions and examples as a plain document and complete the next one (generation stops at the end of the list); post-trained models see the same instructions and examples as earlier chat turns. Invalid samples fall below 0.5% for every model (Table 14).

The examples help base models most. Their pass@1 rises from 0.178 to 0.306 for OLMo-3-1025-7B, from 0.326 to 0.528 for OLMo-3-1125-32B and from 0.500 to 0.647 for Qwen2.5-32B, and their pass@32 reaches 0.76–0.90, while their stated lists barely change (TV 0.712–0.787, against 0.726–0.794 zero-shot). The losses in the samples therefore grow (Table 15): post-training costs 15.2–29.9 points of pass@32 in every pair, including both 32B pairs, whose zero-shot losses did not survive correction, and every post-trained model's Cov@32 (0.21–0.25) is below every base model's (0.33–0.45) and below a uniform guesser's.

For the three Qwen pairs the stated-list improvement persists and mostly grows: TV −0.057 to −0.087, −0.055 to −0.073 with the true pathology removed, and DDF1 +0.051 to +0.081, all surviving correction. OLMo 3 7B remains the exception (TV +0.021, n.s.; +0.024 without the true pathology). For OLMo 3.1 32B the improvement shrinks: the post-trained list still places more reference mass on the conditions it names (+0.047), but

its TV (−0.022), TV without the true pathology (−0.006) and DDF1 (+0.009) no longer survive correction. At 32B the stated-list result is therefore robust for Qwen2.5-32B but depends on the prompt for OLMo 3.1 32B. Post-trained lists still name 5–10 of the 17 severe conditions less often than their base model's.

The consequences replicate in all five pairs. Majority voting over 32 samples gains base models 6.2–10.9 points and post-trained models 0.2–1.1. The three most frequent of 32 sampled answers beat the top three of one stated list by 8.2–22.0 points for base models, while for post-trained models the list is ahead by 0.2–16.7. Agreement with four more samples is the better error signal for every base model (AUROC 0.79–0.84, against 0.67–0.73 for the stated probability), and the stated probability for every post-trained model (0.69–0.77, against 0.56–0.68); against agreement across 31 samples the stated probability wins clearly only for Qwen3-8B and Qwen2.5-7B-Instruct (+0.108 and +0.130, clustered) and ties for the others. This check uses one temperature and one set of examples, and the examples show the reference's own lists, which may help every model match the reference.

F.2. Two further controls

Reference-free few-shot examples. The examples of §F.1 show each training patient's reference top five, and so also the target of the evaluation. We re-ran the six final pairs with the same three patients but with each example list replaced by Qwen2.5-32B-Instruct's own greedy answer (on-list, with the true pathology first for two patients and second for one). The conclusions do not change (Table 16). Post-training costs 12.7–29.7 points of pass@32 in every pair. The stated list moves closer to the reference for the three Qwen pairs and, now clearly, for Mistral (TV −0.053 to −0.077; −0.031 to −0.059 with the true pathology removed), but not for OLMo 3 7B or OLMo 3.1 32B, and wherever the list improves its allocation among the named conditions also moves closer to the reference (conditional TV −0.051 to −0.100). Voting helps base models by 3.7–13.7 points and post-trained ones by at most 1.4; the top three of one list beats the three most frequent samples for every post-trained model; the stated probability beats agreement as an error signal for every post-trained model and agreement beats it for every base model; and no pair names more severe conditions less often than chance. Five of six base models score lower at k=1 than with reference-derived examples (OLMo-3-1125-32B 0.453 against 0.528).

Matched temperature. Post-trained models are sharper than their base models at the same nominal temperature, so part of the narrowing could be colder effective sampling. We sampled Qwen3-8B and Qwen2.5-7B-Instruct at T=1.3 and 1.6 (Table 17), and the two models differ. Qwen3-8B barely broadens: at T=1.6 it gives 1.92 distinct answers and reaches 0.684 at pass@32, still below its base model at T=0.7 (0.732) and at T=1 (0.789; −10.5 points, surviving correction); voting gains stay under 2 points, agreement stays weaker than for the base model, and its stated list does not change (TV 0.701). Qwen2.5-7B-Instruct does broaden: at T=1.3, with 2% invalid answers, it gives 3.38 distinct answers, matches its base model's pass@32 at T=0.7 (0.704 against 0.695), gains 5.6 points from voting, and agreement is again the better error signal (0.794 against 0.602), at a cost of 4.2 points of pass@1; at T=1.6, 31% of its answers are invalid. Its stated list stays closer to the reference than the base model's (0.722 against 0.779). For some models, then, much of the

sampled narrowing is sharper sampling that a higher temperature undoes; the stated-list result is not.

F.3. Greedy decoding, temperature frontier and a must-not-miss prompt

Greedy single call. Deployed systems often decode greedily. We compared one greedy answer per case under three prompt pairings (Table 18): both arms with the reference-derived worked examples of §F.1 (*same examples*); base models with the reference-free examples of Table 16 against zero-shot post-trained models, so that neither arm sees a reference list (*reference-free base*); and reference-example base models against zero-shot post-trained models, the pairing we first ran. Under every pairing post-training significantly raises top-1 in at most two of six pairs. With the same examples these are Qwen3-8B (+13.7) and Mistral (+10.8); Qwen2.5-7B gains +1.8 and the other three lose 4.8–5.8 points, all n.s. With reference-free base models only Qwen3-8B gains (+10.7), while Qwen2.5-32B (−8.0) and OLMo 3 7B (−14.5) lose significantly. The example lists matter little to the base models (top-1 within 2.6 points, except OLMo 3.1 32B, 0.630 against 0.559) but raise five of the six post-trained models (Mistral 0.417 to 0.577, OLMo 3 7B 0.258 to 0.345; Qwen2.5-7B-Instruct falls to 0.458). Under every pairing the two smaller Qwen models' lists contain the true pathology more often (recall@5 +16.8 to +26.0) and name more warranted severe conditions, every Qwen list is closer to the reference, and OLMo 3 7B's lists are farther. Much of post-training's advantage at one sampled call (+12.6 to +27.5 points at T=0.7 and 1, §4.1) is therefore the base model's sampling noise, which greedy decoding also removes.

Temperature frontier. With worked examples and top-p fixed at 1, we sampled the three 7–8B pairs at $T = 0.4$, 0.7, 1.0 and 1.3 (Table 19). Post-trained pass@1 barely moves with temperature (Qwen3-8B 0.60, Qwen2.5-7B-Instruct 0.43–0.46, OLMo-3-7B-Instruct 0.29–0.35), pass@32 rises (to 0.70, 0.66 and 0.74 at T=1.3), and the stated list does not change (TV 0.68, 0.70, 0.81), consistent with the concentration described by Yang et al. (2026). Base models trade pass@1 (Qwen3-8B-Base 0.45 to 0.18) for pass@32, which peaks at T=1 or 1.3 (0.84–0.88), and 25–36% of their answers are invalid at T=1.3. The frontiers differ by family. For Qwen3-8B the post-trained model is better at pass@1 and the base model at pass@32 at every temperature tested; for Qwen2.5-7B the curves meet (at pass@1 about 0.43, the base model at T=0.4 and the Instruct model at T=1.3 both reach pass@32 0.66); and for OLMo 3 7B the base model at T=0.4 beats its Instruct model at every temperature on both (0.368 and 0.803, against at best 0.345 and 0.737). Agreement becomes a better error signal for post-trained models as temperature rises (Qwen3-8B 0.57 to 0.68) but stays below the base models' (0.81–0.92).

Must-not-miss prompt. The standard prompt asks for the most likely conditions. We asked the six post-trained models for the five most likely conditions and then up to three dangerous conditions to rule out (Table 20). Whole lists then name 0.44–0.58 of the severe conditions that the reference weights at least 0.1, 6–15 points more than with the standard prompt and 11–21 more than the base models' lists, and acute pulmonary edema is recovered for every model but Qwen3-8B (e.g. Qwen2.5-32B 0.44 → 0.63, OLMo 3 7B 0.31 → 0.58). The cost is 1.1–2.1 more unwarranted severe conditions per list, mostly pulmonary embolism, named in 88–100% of eligible cases, and for Qwen3-8B, Qwen2.5-7B-Instruct and

Mistral a shorter most-likely part that contains the true pathology 5–10 points less often. Even when asked, no model names more than 0.58 of the warranted severe conditions.

F.4. Must-not-miss conditions defined outside DDXPlus

The severity label used above is the dataset's own. We therefore re-ran every severe-condition analysis with two published sets of conditions that must not be missed, mapped onto DDXPlus's 49 (Table 21): a harm-based set, the DDXPlus conditions among the 15 diseases that cause about half of serious misdiagnosis-related harms in the USA (Newman-Toker et al., 2024) (NSTEMI/STEMI, pulmonary embolism, pneumonia and pulmonary neoplasm, of which DDXPlus labels only the first two severe), and the life-threatening causes of acute chest pain in the 2021 AHA/ACC guideline (Gulati et al., 2021) (acute coronary syndrome as NSTEMI/STEMI and unstable angina, pulmonary embolism, esophageal rupture as Boerhaave, and pneumothorax). Their union has seven conditions. Neither set involves a judgment of these vignettes, and both were fixed before the analysis. On average the result is stronger than with DDXPlus's labels. Zero-shot, post-trained lists name 9–24 points more of the warranted union conditions in seven of eight settings (OLMo 3.1 32B −0.3, n.s.), and with worked examples 7–22 points more for the three Qwen models and OLMo 3 7B (OLMo 3.1 32B +6.5 and Mistral +4.0, n.s.); the harm-based and chest-pain sets agree. The losses of specific conditions, however, are confined to OLMo, Mistral and the few-shot Qwen2.5 models. No Qwen model names any union condition significantly less often zero-shot, while OLMo 3 7B loses NSTEMI/STEMI and unstable angina at both temperatures, OLMo 3.1 32B loses NSTEMI/STEMI, pulmonary embolism and pneumothorax, and with worked examples Mistral loses four of the seven (pulmonary embolism 0.31 points), Qwen2.5-7B-Instruct two and Qwen2.5-32B-Instruct one (Boerhaave). When the true pathology is itself in the union set (140 cases), zero-shot post-trained lists contain it among their top five 10–26 points more often for four of five pairs (OLMo 3 7B +9.9, n.s.). Scored against the union set, the must-not-miss prompt names 15–26 points more of the warranted conditions than the standard prompt and 8–20 points more than two combined standard lists, a larger share of its gain than under DDXPlus's labels, at a cost of 0.7–1.5 more unwarranted union conditions per list.

F.5. A third family: Mistral-7B v0.3

We repeat the comparison with Mistral-7B-v0.3 and Mistral-7B-Instruct-v0.3, a family also studied by Jang et al. (2026). With the zero-shot prompt the base model does not answer: 52% of its samples at T=0.7 and 76% at T=1 are invalid (it continues the document instead), and its valid answers mostly repeat the first listed condition, so its single-call accuracy is 6% and the zero-shot pair is not informative. With the few-shot prompt of §F.1 the base model answers validly (0.4% invalid), and the pair reproduces the other families (Table 22). Post-training raises pass@1 by 14.0 points and lowers pass@32 by 14.5, cuts distinct answers from 5.0 to 1.3 and Cov@32 by 0.138, and moves the samples away from the reference (TV +0.058). Voting gains the base model 5.6 points and the Instruct model 0.8; the three most frequent of 32 samples beat one list by 12.6 points for the base model, while for the Instruct model the list is ahead by 7.0; agreement is the better error signal for the base model (AUROC 0.826 against 0.671) and the stated probability for the Instruct model

(0.747 against 0.613). The stated list improves modestly (TV −0.036 and DDF1 +0.046, surviving clustering and Holm correction over 42 tests; −0.023 with the true pathology removed, which does not), the smallest significant gain of the three families, and 11 of the 17 severe conditions are named less often. Mistral-Instruct also names 0.4 conditions per list that are not among the 49.

Appendix G. Thinking mode, aggregation and the error signal

G.1. Thinking mode

Qwen3-8B can reason in a thinking block before it answers; the main runs turn this off. With thinking on (zero-shot prompt, T=0.7, up to 8,192 tokens; traces average about 1,500 tokens and all end before the limit), the same weights behave differently (Table 22). Accuracy rises at both ends: pass@1 is 0.671 against 0.559 without thinking, and pass@32 is 0.815, above the non-thinking model (0.607) and level with the base model (0.732 zero-shot, +8.4 [1.7, 15.3] points, not surviving Holm correction over 42 tests; 0.809 with worked examples). The samples are still narrow (1.76 distinct answers against 3.77 for the base model; Cov@32 0.283 against 0.323) but no farther from the reference than the base model's (TV 0.745 against 0.750). The stated list is the closest of any 7–8B model (TV 0.670, against 0.700 without thinking). The shift in error signals reverses: agreement across 31 samples is again the better signal (AUROC 0.732 against 0.638; difference 0.094 [0.027, 0.170]), as for base models, and voting gains 1.5 points (n.s.). The narrowing of the samples and the advantage of the stated probability are therefore properties of answering directly, which a reasoning mode largely undoes, consistent with the chain-of-thought prompt of §D.3.

G.2. Averaging stated lists across calls

Stated lists can be averaged across calls, as samples can be voted. For each case we average the first 5 or 32 valid stated lists, each renormalized over the conditions it names, and score the mean against the reference (Table 23). Averaging helps base models far more than post-trained ones, whose lists vary little between calls (at T=0.7 with the zero-shot prompt, 32 averaged lists name 7.6–13.4 conditions, against 16.6–22.0 for base models). The mean of five base-model lists is about as close to the reference as one post-trained list (Qwen3-8B 0.699 against 0.700, Qwen2.5-7B 0.717 against 0.717, OLMo 3.1 32B 0.691 against 0.701). With five lists on both sides, the post-trained mean is closer for Qwen2.5-7B under both prompts (−0.025 and −0.027) and for Qwen2.5-32B and Qwen3-8B under one prompt each, not significantly closer for OLMo 3.1 32B (zero-shot) and Mistral (few-shot), and farther for OLMo 3 7B and few-shot OLMo 3.1 32B. Averaged base lists also name severe conditions at least as often, and more often for OLMo 3.1 32B, OLMo 3 7B, Mistral and few-shot Qwen2.5-32B (by 0.06–0.14). One post-trained call thus gets about as close to the reference as several base calls; the packing test (Table 24) shows that it is a stabilized version of the base model's view of the patient rather than its literal average, and, as for the open-ended questions of Lake et al. (2025), post-training adds no information that averaged base lists lack. Qwen3-8B's thinking mode goes furthest: five of its lists beat five base-model lists by more than any direct-answer model's (TV −0.049) and name more severe conditions (+0.129). For a post-trained model answering directly, averaging five lists instead of reading

one improves TV by only 0.005–0.025 at T=0.7 (up to 0.038 at T=1); it names 6–17 points more severe conditions, mostly by naming one to three more conditions. Five post-trained lists still cover 0.44–0.50 of the reference mass at T=0.7 (OLMo 3 7B 0.37) with six to eight conditions, about twice what their 32 samples cover (0.22–0.28) and as much as a uniform guesser naming about 24 conditions (0.48), so the coverage deficit of §4.1 is one of the sampled readout.

G.3. Which signal to use: combination and held-out audit

Whether the stated probability or agreement across samples is the better error signal varies across models, so we ask what predicts it and whether a practitioner can tell. We pool 56 model–setting combinations scored on DDXPlus: the base/post-trained pairs at both temperatures and with three prompts, the OLMo stages, the 32B models, the medical models, our GRPO runs, and the full-distribution prompt. They share weights, prompts and temperatures, so we treat the correlations below as descriptive. The stated probability's advantage over five-sample agreement correlates with the *single-answer rate*, the share of cases whose five samples all give the same top answer (Spearman 0.75), with the stated probability's own AUROC (0.82; the gap contains this term, so part of this is by construction), and negatively with the number of distinct answers (−0.74). The stated probability is the better signal for 19 of the 56. No single-answer threshold separates them: Qwen2.5-7B-Instruct still favors agreement, while OLMo 3 7B's DPO and Instruct models favor their unusually informative stated probability. With five-sample agreement, as in the original audit, combining the signals beats choosing between them; with agreement across all 31 further samples the advantage shrinks (post-trained combinations: agreement 0.742, rank average 0.761, +0.019 $[-0.007, 0.040]$; logistic 0.772, +0.030 $[0.013, 0.044]$), and for the 17 base combinations agreement alone (0.821) beats the rank average (0.780) and ties the logistic combination (0.827). The rest of this section uses five-sample agreement. Fitting a logistic model of correctness on the stated probability and five-sample agreement with 100 labeled cases, and evaluating on the other 880, gives a mean held-out AUROC of 0.753 for the 39 post-trained combinations (0.765 for all 56); averaging the two signals' ranks without fitting gives 0.746; picking the better single signal on the same 100 cases gives 0.720; always using agreement 0.695; always using the stated probability 0.671. The logistic combination beats the single-signal pick for 35 of the 39 post-trained combinations. This matches the general finding that hybrids of consistency and verbalized confidence beat either (Xiong et al., 2024). For choosing a single signal, the simplest rule, always using agreement, is right for 66% of the combinations (54% of the 39 post-trained ones) and loses 0.025 AUROC on average (0.032). An audit does better: choose the signal with the higher AUROC on 100 random labeled cases, using five samples per case, and evaluate the choice on the other 880. Over 500 draws per combination it picks the better held-out signal 86% of the time (82% for post-trained models) and loses 0.005 AUROC (0.007); with 50 cases, 82% and 0.010; with 200, 90% and 0.003. It is reliable when the two signals differ by more than 0.05 AUROC (96% of draws, 39 combinations) and close to a coin flip when they differ by less than 0.02 (53%, 5 combinations), where the choice matters least. The audit's main cost is gold labels for 100 cases from the target population, and we validate it only within DDXPlus.

Appendix H. Medical post-training and rare disease

H.1. Medical post-training

Applied after instruction tuning, medical RL does not cost coverage. AlphaMed improves on its parent at every k (Figure 4): +3.9 points at k=1 ([+2.2, +5.5]; clustered [−0.4, +8.4]), widening to +8.4 at k=32 ([+6.0, +10.6]; clustered [+2.6, +14.7]). It reaches at least one correct sample for 47 of 49 conditions, against 39 for its parent. It does not restore breadth, though: distinct answers (1.87 vs. 1.82), entropy (0.28 vs. 0.30), and coverage (0.27 vs. 0.24) stay at the instruction-tuned level, far below the base model. AlphaMed also names off-list conditions more often than its parent, a mild loss of instruction following. By condition, AlphaMed slightly *raises* consideration in every severity tier (+2.2 to +4.3 points; Table 6). It still loses acute pulmonary edema (0.31 to 0.13), and it introduces an attractor of its own: spontaneous pneumothorax as the top answer in 10.3% of samples, against 4.3% for its parent and a reference rate of 1.0%. Medical post-training thus operates on an already collapsed distribution; the collapse comes from the general post-training stage. The other two descendants go further (Table 27). HuatuoGPT-o1 (SFT then PPO) and m1 (SFT distillation from a reasoning teacher) both *widen* the differential relative to their parent: 2.49 and 3.29 distinct answers vs. 1.82, and coverage@32 of 0.31 and 0.33 vs. 0.24. Their pass@32 rises by 13.7 [+11.3, +16.0] and 18.1 [+15.5, +20.7] points (clustered [8.0, 20.0] and [10.5, 25.9]). m1 gains almost nothing at k=1 (+0.9, interval includes zero), so its improvement lies entirely in breadth. Both reasoning-trained models are also less calibrated (ECE 0.17 and 0.21 vs. 0.11). Medical post-training on an instruction-tuned model thus never adds to the collapse, and reasoning distillation partially reverses it.

Medical RL from the base model. Alongside the instruct-initialized AlphaMed-7B, the AlphaMed authors released a companion checkpoint, AlphaMed-7B-base-rl, initialized from Qwen2.5-7B rather than Qwen2.5-7B-Instruct (according to its configuration file). The paper (Liu et al., 2025) describes only the instruct-initialized models: GRPO with a binary correctness reward on about 19k multiple-choice questions from MedQA and MedMCQA, with a 64-question by 8-sample batch for 300 steps. It does not document the base-initialized checkpoint, so we cannot rule out differences in data or hyperparameters, and we treat it as corroborating rather than controlled evidence. It offers a third-party check on whether RL itself causes the collapse (Table 27, lower block). It does not. RL from the base model improves on the base at every k (+14.3 [+13.1, +15.7] points at k=1, +9.5 [+7.4, +11.6] at k=32). It keeps 3.14 distinct answers, against 1.87 for the instruct-initialized AlphaMed. Against that instruct-model AlphaMed, it is 2.2 [0.4, 3.9] points worse at k=1 and 10.8 [+8.6, +13.2] points better at k=32. Its severe-tier Considered@32 shows no detectable change relative to the base (−0.7 [−3.0, +1.7] points), against −12.4 for instruction tuning and −8.1 for instruction tuning followed by AlphaMed's RL (Table 6; all well below the chance levels of §D.2). RL from a base model also narrows less in OLMo's RL-Zero at T=1 and in our GRPO runs (§I), but not uniformly: RL-Zero at T=0.7 loses about as much as SFT, and within OLMo 3 7B's instruct pipeline the RL stage after SFT and DPO is the largest single narrowing (§D.4).

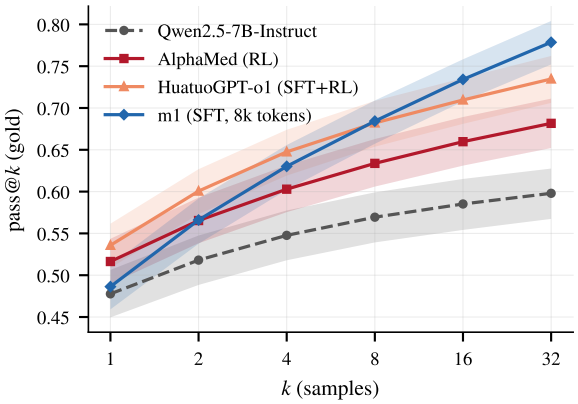


Figure 4: DDXPlus at T=0.7 (980 cases): Qwen2.5-7B-Instruct (dashed) vs. its medical descendants, with 95% case-bootstrap bands. Every descendant matches or beats its parent at every k, and the gaps widen with k: there is no sharpening tax at the medical stage.

H.2. Rare disease: overconfidence without accuracy

On 1,401 Phenopacket Store cases covering 749 diseases, every 7–8B model is near the floor: pass@1 is 1.4–2.2% and pass@32 3.7–7.0% for the Qwen-family models, and lower for OLMo 3 7B (2.5–4.5%). Stated confidence is not near the floor. Qwen2.5-7B-Instruct states a mean top-1 probability of 0.51, and its RL-trained medical descendants state 0.67 (HuatuoGPT-o1) and 0.69 (AlphaMed), while their accuracy is about 2%. ECE accordingly rises from 0.49 to 0.64–0.67; at this accuracy ECE mostly restates mean confidence, so we read it as a rise in confidence without a matching rise in accuracy. The RL-trained medical models (AlphaMed; HuatuoGPT-o1 after SFT) thus state much higher confidence where they are least accurate. The effect also appears without instruction tuning: AlphaMed's base-initialized checkpoint is the most overconfident Qwen-family model (mean confidence 0.71, ECE 0.70, accuracy 1.7%). The medical descendants do gain a little accuracy: AlphaMed beats its parent at every k (+0.6 [+0.2, +0.9] points at k=1, +1.0 [+0.1, +2.1] at k=32), HuatuoGPT-o1 at $k \leq 8$, and m1 at $k \geq 2$ (+1.4 [+0.4, +2.5] at k=32). These gains are an order of magnitude smaller than the rise in confidence. m1, trained only with supervised reasoning traces (at its intended 8k-token budget), is the exception on calibration and diversity: it is the most diverse model (15.3 distinct answers per 32 samples, against 11.5 for the parent and 9.2 for HuatuoGPT-o1), and its confidence rises least (0.57).

General post-training reproduces the DDXPlus pattern on Qwen3. Qwen3-8B beats its base at pass@1 (+0.7 [+0.4, +1.1]) and trails it from k=8 on, by −2.2 [−3.1, −1.3] points at pass@32 (a relative loss of 37%), with half the distinct answers (6.4 vs. 13.4). For Qwen2.5 no pass@k difference is significant. Unlike medical RL, general post-training *lowers* stated confidence here (0.61 to 0.46 for Qwen3, 0.61 to 0.51 for Qwen2.5), although base models also give invalid answers more often (8.5% of Qwen3-8B-Base samples). OLMo 3 does not share this: every stage of its pipeline states a mean confidence of 0.73–0.84 at about 1% accuracy (pass@32 2.5–4.5%), and its SFT checkpoint is the most overconfident model we

evaluate (confidence 0.84, ECE 0.83). On the 500-case subset, our 8B GRPO runs (§I) are indistinguishable from their base (all pass@k differences within ±0.6 points).

Accuracy depends strongly on when the source case report was published (Table 28). For every model, pass@32 is 5–10% on cases published up to 2022 (n=901), at most 1.7% on 2023–24 cases (n=348), and 2.6–4.6% on 2025+ cases (n=152). The pattern is not monotone, since 2025+ cases score above 2023–24 cases, so case mix also matters; the gap between older and recent cases is consistent with, but does not establish, memorization of older case reports.

H.3. Stated and sampled readouts on real rare-disease cases

Phenopacket Store cases are real, but they have no reference differential and accuracy is near the floor, so we can test only the consequences of §4.3, and we count an answer as correct if it is the diagnosis or a closely related Mondo class (exact matches give only 8–31 correct single-call answers per model). Top answers are grouped by Mondo class for voting and agreement (Table 29). The DDXPlus shifts reappear, but the top-three comparison does not flip as it does on DDXPlus. Majority voting over 32 samples helps the base models by 3.1–4.6 points but the final post-trained models by only 0.4–1.5 (Qwen3-8B +0.4, n.s.). Post-training weakens agreement as an error signal (Qwen3-8B 0.830 → 0.763, Qwen2.5-7B 0.845 → 0.745, OLMo 3 7B 0.814 → 0.726) and strengthens the stated probability (0.567 → 0.707, 0.598 → 0.717, 0.464 → 0.652). Agreement beats the stated probability by 0.25–0.35 AUROC for the base models and by only 0.03–0.07 for the post-trained ones, significantly only for Qwen3-8B (-0.056 $[-0.120, -0.002]$). On these open-ended cases the three most frequent of 32 sampled answers still contain a correct diagnosis more often than one list's top three, but the gap shrinks from 4.6–7.0 points for the base models to 2.7–3.6 for Qwen2.5-7B-Instruct and OLMo-3-7B-Instruct and to a tie for Qwen3-8B ($+0.4$ $[-0.8, +1.8]$). The medical models behave like their post-trained parent, with agreement the better signal for all of them.

Appendix I. GRPO with a group-level reward

We also fine-tuned Qwen3-1.7B (instruct) and Qwen3-8B-Base with GRPO (LoRA, 200 steps, 16 samples per prompt, KL coefficient 0.04) on DDXPlus training patients disjoint from the test split, comparing the standard correctness reward (`acc`) with a group-level reward (`gm`). If k_a of the n samples in a GRPO group give top answer a, `gm` adds $\mathrm{clip}(n\,p_{\mathrm{ref}}(a) - (k_a - 1), 0, 1)$, which pays an answer only while its condition is under-represented relative to its reference mass. At 1.7B, a boosted variant (weight 3, training temperature 1.2, so the two changes are confounded) raised pass@1 by 1.6 and distinct answers by 0.10 over `acc` (two seeds; intervals ignore seed variation). From Qwen3-8B-Base, both recipes reached nearly the official Qwen3-8B's top-1 accuracy (0.537 and 0.539 vs. 0.558) while keeping more than twice its distinct answers (2.7–2.9 vs. 1.27), and `gm` narrowed the base model less than `acc` (distinct answers +0.14 at equal pass@1). Neither changed Phenopacket accuracy. Because `gm` needs a reference differential at training time, which real clinical data rarely provide, and the runs train on the evaluation distribution, these are existence results only. Closely related rewards computed from group answer counts have appeared since: a witness advantage that trains models to sample the distributions they

state (Arabi et al., 2026), and a batch bonus that penalizes repeated answers (Song et al., 2025); we do not compare against them. The GRPO runs enter only the 56-combination audit (§G.3).

Appendix J. Condition-clustered intervals and multiple comparisons

DDXPlus has 20 test cases for each of 49 true pathologies, and effects can be specific to a condition, so cases are not independent. Table 30 gives the headline base-vs-post differences with a bootstrap that resamples the 49 pathologies with all their cases (50,000 resamples for these headline tests, so that results near the threshold do not flip between runs; 10,000 elsewhere). Two-sided p-values use the add-one correction, $2(1 + \min(n_{\leq 0}, n_{\geq 0}))/(B + 1)$, so the smallest attainable value lies below the strictest Holm threshold, 0.05/112. Holm correction runs over the 112 tests (8 pairs $\times$ 14 metrics fixed in advance: pass@1, pass@32, distinct answers, coverage@32, sampled TV, list TV, mass named, severe named, list and sample TV without the true pathology, DDF1, unwarranted severe conditions named, and the two AUROCs; the table shows seven). The same procedure is applied separately to the few-shot pairs (70 tests), the Mistral and thinking-mode pairs (42) and the stage and medical comparisons (154). Not surviving correction among the headline tests shown, marked †: list TV without the true pathology for Qwen3-8B at T=0.7; sampled TV for Qwen2.5-7B at T=0.7; sampled TV for Qwen2.5-7B at T=1; pass@32 for Qwen2.5-32B at T=0.7; list TV for OLMo 3 7B at T=0.7; DDF1 for OLMo 3 7B at T=0.7; agreement AUROC for OLMo 3 7B at T=0.7; list TV for OLMo 3 7B at T=1; list TV without the true pathology for OLMo 3 7B at T=1; agreement AUROC for OLMo 3 7B at T=1; pass@32 for OLMo 3.1 32B at T=0.7; sampled TV for OLMo 3.1 32B at T=0.7; agreement AUROC for OLMo 3.1 32B at T=0.7. OLMo 3 7B's list TV without the true pathology at T=0.7 survives as a significant *increase*.

Claim	Primary evidence	Holds under	Weaker or fails under
(a) Post-training keeps the base model's view of each patient and removes its variation between calls (§4.2)	A post-trained list is closer to its own base model's mean list for the same patient than to the means for other patients with the same diagnosis, in all 15 settings (TV −0.028 to −0.076); post-trained lists vary 0.10–0.33 between calls, base lists 0.37–0.69 (Table 24)	worked examples; thinking mode; against other families' base models (−0.03 to −0.14)	within a family; OLMo 3 7B drifts from its base model's view
(b) Sampled answers collapse (§4.1)	pass@32 −9.7 to −14.6 points at 7–8B zero-shot; 1.3–2.4 distinct answers against 3.8–4.9 (Table 1)	worked examples of either kind (13–30 points in all six pairs); Mistral; OLMo stages; 32B with worked examples	32B zero-shot (n.s.); Qwen2.5-7B-Instruct at T=1.3; thinking mode; RL-Zero at T=1
(c) Post-training's first-answer gain is mostly sampling noise (§4.1)	Decoded greedily, post-training raises top-1 significantly in at most two of six pairs under either matched prompt (Table 18)	both matched pairings; both 32B pairs	Qwen3-8B (+10.7 and +13.7); Mistral with shared examples (+10.8)
(d) One stated list improves robustly only for Qwen (§4.2)	Recall@5 +12 to +24 points, TV −0.043 to −0.088, conditional TV −0.025 to −0.078 (Table 2, Table 30)	complete lists; correct and wrong top answers; true pathology removed; worked examples for Qwen3-8B and Qwen2.5-7B; thinking mode	OLMo 3 7B (no gain); OLMo 3.1 32B and Qwen2.5-32B recall with worked examples; five averaged base lists match one post-trained list
(e) Severe conditions are named more often on average, but specific ones are lost (§4.2)	Named more often in every zero-shot setting (+2.2 to +13.8 points, six of eight surviving correction); 2–7 of 17 conditions per pair named significantly less often (Table 11)	published must-not-miss sets (+9 to +24 points in seven of eight settings)	worked examples (mean falls for Qwen2.5-32B and OLMo 3 7B); averaged lists; with published sets only OLMo, Mistral and few-shot Qwen2.5 lose specific conditions
(f) Asking for must-not-miss conditions raises coverage, partly by lengthening the list (§4.2)	+6 to +15 points of warranted severe conditions named, 0.02–0.11 beyond two combined standard lists (Table 20)	published must-not-miss sets (+15 to +26 points; 8–20 beyond two lists)	costs one to two more unwarranted severe conditions per list
(g) Voting stops helping and agreement weakens as an error signal (§4.3)	Voting +3.4 to +8.4 points for base models, at most +1.9 for post-trained ones; agreement's AUROC falls in every family (Table 26)	worked examples; Mistral; rare-disease cases	models that reason first; thinking mode; Qwen2.5-7B-Instruct at T=1.3; agreement drop n.s. for OLMo
(h) The better error signal depends on model and prompt; combining helps post-trained models modestly (§4.3)	Held-out audit over 56 model–setting combinations: logistic combination +0.030 [0.013, 0.044] over 31-sample agreement for post-trained models, rank average +0.019 (n.s.) (§G.3)	worked examples (stated probability better for three post-trained models)	base models and thinking mode (agreement alone as good or better); rare-disease cases (agreement better)

Table 3: The claims of the paper and the evidence for them. Ranges are post-trained minus base unless stated; "n.s.": the condition-clustered interval includes zero or the test does not survive correction.

Model	Parent	Post-training	Role
Qwen2.5-7B	n/a	none (pretrained)	base
Qwen2.5-7B-Instruct	Qwen2.5-7B	general SFT + RLHF	instruct; medical parent
AlphaMed-7B (Liu et al., 2025)	Qwen2.5-7B-Instruct	rule-based RL	medical
HuatuoGPT-o1-7B (Chen et al., 2025)	Qwen2.5-7B-Instruct	SFT + PPO (verifier)	medical
m1-7B-23K (Huang et al., 2026)	Qwen2.5-7B-Instruct	SFT distillation	medical
Qwen3-8B-Base	n/a	none (pretrained)	base
Qwen3-8B	Qwen3-8B-Base	general post-training	instruct
Qwen2.5-32B	n/a	none (pretrained)	base
Qwen2.5-32B-Instruct	Qwen2.5-32B	general SFT + RLHF	instruct
OLMo-3-1025-7B (Team Olmo et al., 2025)	n/a	none (pretrained)	base
OLMo-3-7B-Instruct-SFT	OLMo-3-1025-7B	SFT	stage 1
OLMo-3-7B-Instruct-DPO	OLMo-3 SFT	DPO	stage 2
OLMo-3-7B-Instruct	OLMo-3 DPO	RLVR	stage 3 (instruct)
OLMo-3-7B-RL-Zero-General	OLMo-3-1025-7B	RL (LM judge)	RL from base
OLMo-3-1125-32B	n/a	none (pretrained)	base
OLMo-3.1-32B-Instruct-SFT	OLMo-3-1125-32B	SFT	stage 1
OLMo-3.1-32B-Instruct-DPO	OLMo-3.1 SFT	DPO	stage 2
OLMo-3.1-32B-Instruct	OLMo-3.1 DPO	RLVR	stage 3 (instruct)

Table 4: Models studied. Each pair or family shares its pretrained weights, so differences reflect post-training. OLMo 3 and OLMo 3.1 release every stage of their instruct pipelines.

Pair	Model	pass@1	pass@32	Distinct	Entropy	TV↓	Cov@32	ECE↓
Qwen3-8B, T=1	base	0.282	**0.789**	**6.60**	**1.31**	**0.712**	**0.407**	0.214
	post-trained	**0.556**	0.643	1.53	0.18	0.783	0.236	**0.061**
Qwen2.5-7B, T=1	base	0.239	**0.766**	**8.23**	**1.58**	**0.715**	**0.422**	0.303
	instruct	**0.464**	0.663	2.65	0.49	0.761	0.290	**0.137**
Qwen3-8B, T=0.7	base	0.388	**0.730**	**3.76**	**0.78**	**0.750**	**0.322**	0.121
	post-trained	**0.558**	0.607	1.27	0.11	0.793	0.217	**0.063**

Table 5: DDXPlus, 32 samples per case (980 cases). The reference differential has mean entropy 1.87. Distinct and entropy refer to the top-1 answers across samples; Cov@32 is the share of reference probability mass covered by those answers. ECE is for the stated top-1 probability.

Comparison	Considered@32 (base → post-trained) Severe (1–2)	Moderate (3)	Mild (4–5)
Qwen3-8B, T=1	0.45 → 0.27	0.50 → 0.32	0.45 → 0.20
	−18.4 [−20.8, −15.9]	−18.1 [−20.7, −15.7]	−24.2 [−27.0, −21.4]
Qwen2.5-7B, T=1	0.51 → 0.34	0.50 → 0.37	0.38 → 0.23
	−17.0 [−19.7, −14.2]	−12.1 [−14.7, −9.7]	−15.0 [−17.4, −12.6]
Qwen3-8B, T=0.7	0.34 → 0.24	0.44 → 0.29	0.31 → 0.18
	−9.8 [−11.8, −7.9]	−15.8 [−18.2, −13.5]	−13.4 [−15.8, −11.1]
Qwen2.5-7B, T=0.7	0.40 → 0.28	0.41 → 0.32	0.28 → 0.17
	−12.4 [−14.7, −10.2]	−9.1 [−11.2, −6.7]	−10.3 [−12.3, −8.3]
Instruct → AlphaMed, T=0.7	0.28 → 0.32	0.32 → 0.34	0.17 → 0.21
	+4.3 [+2.0, +6.6]	+2.2 [+0.4, +4.0]	+3.9 [+2.4, +5.4]
Instruct → HuatuoGPT-o1, T=0.7	0.28 → 0.36	0.32 → 0.37	0.17 → 0.27
	+8.4 [+6.1, +10.8]	+6.0 [+4.0, +7.9]	+9.5 [+7.5, +11.5]
Instruct → m1, T=0.7	0.28 → 0.41	0.32 → 0.41	0.17 → 0.29
	+13.5 [+10.8, +16.2]	+9.7 [+7.2, +12.2]	+11.4 [+9.2, +13.6]
Base → AlphaMed (base-RL), T=0.7	0.40 → 0.39	0.41 → 0.40	0.28 → 0.27
	−0.7 [−3.0, +1.7]	−0.4 [−2.6, +1.7]	−0.4 [−2.4, +1.7]
Base → `acc` GRPO (8B), T=0.7	0.34 → 0.31	0.44 → 0.42	0.31 → 0.27
	−2.8 [−4.4, −1.3]	−2.1 [−3.7, −0.6]	−4.3 [−6.2, −2.5]
Base → `gm` GRPO (8B), T=0.7	0.34 → 0.32	0.44 → 0.43	0.31 → 0.28
	−1.5 [−3.2, +0.1]	−1.3 [−3.0, +0.3]	−3.0 [−4.8, −1.2]

Table 6: Share of (case, condition) pairs with reference probability ≥ 0.1 in which the condition is a top-1 answer in at least one of 32 samples, by DDXPlus severity. Second rows give differences in points with 95% case-bootstrap intervals. All rows use the 980 test cases; GRPO rows are seed 0. General post-training lowers consideration in every tier, including the severity-1–2 tier; medical post-training raises it; our GRPO from the base model loses at most 4.3 points, and with the group reward the severe-tier loss is not significant.

		pass@1		pass@32		Distinct	
Prompt	T	Base	Post	Base	Post	Base	Post
default	0.7	0.388	**0.558**	**0.730**	0.607	3.76	1.27
default	1.0	0.282	**0.556**	**0.789**	0.643	6.60	1.53
plain	0.7	0.352	**0.583**	**0.727**	0.655	3.72	1.28
plain	1.0	0.244	**0.586**	**0.792**	0.705	5.57	1.54
CoT	0.7	0.409	**0.652**	0.788	**0.835**	3.89	2.08
CoT	1.0	0.275	**0.642**	0.837	0.837	6.31	2.24

Table 7: Qwen3-8B-Base vs. Qwen3-8B on DDXPlus (980 cases, 32 samples) under three prompts. Bold marks the significantly better model (paired case bootstrap). Without reasoning, the base model wins at k=32; with CoT, the post-trained model matches or beats it, but its samples stay far less diverse.

T	Stage	pass@1	pass@32	Distinct	Entropy	Cov@32	Severe C@32
0.7	Base	0.178	**0.705**	**4.93**	**1.16**	**0.351**	**0.412**
	SFT	0.222	0.559	4.01	0.87	0.257	0.373
	SFT+DPO	**0.324**	0.613	3.30	0.63	0.272	0.338
	SFT+DPO+RLVR (Instruct)	0.266	0.422	2.37	0.42	0.195	0.251
	RL-Zero (RL from base)	0.295	0.572	4.06	0.84	0.286	0.326
1.0	Base	0.128	**0.713**	**7.53**	**1.66**	**0.398**	**0.469**
	SFT	0.191	0.634	6.32	1.29	0.321	0.426
	SFT+DPO	**0.313**	0.688	4.48	0.87	0.312	0.410
	SFT+DPO+RLVR (Instruct)	0.272	0.561	3.58	0.68	0.259	0.325
	RL-Zero (RL from base)	0.279	0.695	6.28	1.21	0.367	0.429

Table 8: OLMo 3 7B post-training stages on DDXPlus (980 cases, 32 samples). Severe C@32 is severe-tier Considered@32 as in Table 6. Bold marks the best value in each block. Diversity falls at every stage of the instruct pipeline, and the RLVR stage loses the most coverage; RL from the base keeps the base model's pass@32 at T=1.

Model	pass@1	pass@32	Distinct	Cov@32	Severe C@32	TV one list↓	TV samples↓	Vote@32 gain
Qwen2.5-32B (base)	0.500	**0.795**	**3.78**	**0.352**	**0.35**	0.726	**0.726**	+7.0
Qwen2.5-32B-Instruct	**0.604**	0.746	1.68	0.276	0.29	**0.674**	0.759	+1.9
OLMo-3-1125-32B (base)	0.326	**0.796**	**4.32**	**0.347**	**0.39**	0.746	**0.742**	+8.4
SFT	0.461	0.674	2.94	0.265	0.27	0.736	0.776	+3.6
SFT+DPO	**0.584**	0.784	2.29	0.293	0.32	**0.694**	0.755	+2.1
SFT+DPO+RLVR (Instruct)	0.560	0.756	2.15	0.283	0.30	0.701	0.763	+0.9

Table 9: 32B replication on DDXPlus (T=0.7, 980 cases, 32 samples). TV columns: distance to the reference differential of one stated list and of the sampled top-1 answers (§4.2). Vote@32 gain: accuracy of the majority answer over 32 samples minus single-call accuracy, in points (§4.3). Bold: best value per family.

	TV to reference↓		AUROC, single-call answer correct			Top-3: one
Model	One distribution	Samples	Stated p	Agree@5	Agree@32	dist. − 32 samples
Qwen3-8B-Base	0.733	**0.752**	0.647	0.809	**0.845**	−6.7
Qwen3-8B	**0.681**	0.790	**0.717**	0.609	0.647	+14.6
Qwen2.5-7B	0.769	**0.758**	0.688	0.792	**0.827**	−10.2
Qwen2.5-7B-Instruct	**0.702**	0.770	0.674	0.715	**0.748**	+1.6
OLMo-3-1025-7B	0.807	**0.766**	0.610	0.744	**0.786**	−15.3
OLMo-3-7B-Instruct	**0.775**	0.787	0.749	0.705	**0.759**	−9.5

Table 10: Full-distribution prompt on DDXPlus (T=0.7, 980 cases): a probability on every condition, top answer = highest stated probability. TV: one sample's stated distribution vs. the sampled top-1 answers (bold: better model of each pair). AUROC: bold marks each model's best signal.

Pair (base → post)	Named less / more often	Significant less / more	P(as many less often)	Mean shift (17 conditions)	P	Largest loss (shift)
Qwen3-8B, T=0.7	5 / 12	2 / 3	0.98	+0.040	0.001	acute pulmonary edema (-0.14)
Qwen3-8B, T=1	6 / 11	2 / 4	0.93	+0.060	0.001	stable angina (-0.06)
Qwen2.5-7B, T=0.7	5 / 12	2 / 3	0.98	+0.029	0.001	stable angina (-0.16)
Qwen2.5-7B, T=1	4 / 13	2 / 5	0.99	+0.065	0.001	stable angina (-0.13)
Qwen2.5-32B, T=0.7	6 / 11	3 / 5	0.93	+0.047	0.001	acute pulmonary edema (-0.19)
OLMo 3 7B, T=0.7	8 / 9	7 / 6	0.71	+0.046	0.001	acute pulmonary edema (-0.24)
OLMo 3 7B, T=1	8 / 9	6 / 6	0.69	+0.061	0.001	NSTEMI/STEMI (-0.12)
OLMo 3.1 32B, T=0.7	7 / 10	3 / 0	0.82	+0.001	0.854	acute pulmonary edema (-0.16)
Qwen3-8B, few-shot	5 / 12	1 / 4	0.97	+0.061	0.001	acute pulmonary edema (-0.16)
Qwen2.5-7B, few-shot	8 / 9	4 / 2	0.69	+0.034	0.001	stable angina (-0.15)
Qwen2.5-32B, few-shot	10 / 7	5 / 3	0.29	−0.012	0.009	laryngospasm (-0.25)
OLMo 3 7B, few-shot	9 / 8	7 / 3	0.51	−0.018	0.018	croup (-0.51)
OLMo 3.1 32B, few-shot	9 / 8	4 / 2	0.50	+0.011	0.012	stable angina (-0.25)
Mistral-7B, few-shot	11 / 6	3 / 0	0.17	−0.009	0.052	pulmonary embolism (-0.31)

Table 11: The 17 severe conditions one by one (DDXPlus). For each condition, the share of a model's lists that name it, over the cases whose reference gives it any weight; post-trained minus base, with intervals that resample the true pathologies (10,000 resamples) and Benjamini–Hochberg correction across the 17 conditions. P columns compare the count of conditions named less often and the mean shift across conditions with a null that swaps each case's base and post-trained lists at random (2,000 swaps); the count test is weak against an overall upward shift, so the mean shift and the largest single loss are shown too.

Model (T=0.7)	Complete lists	TV of one list↓ complete only	top-1 right	top-1 wrong	Excess TV over floor↓ One list	Samples	Severe named ref. < 0.1	pass@32 list-sampled
Qwen3-8B-Base	0.90	0.743	0.663	0.806	0.479	**0.557**	0.752	0.780
Qwen3-8B	0.99	**0.701**	**0.646**	**0.769**	**0.435**	0.613	**0.680**	**0.811**
Qwen2.5-7B	0.95	0.779	0.689	0.828	0.514	**0.575**	0.823	0.772
Qwen2.5-7B-Instruct	0.91	**0.719**	**0.655**	**0.773**	**0.452**	0.605	**0.796**	**0.802**
Qwen2.5-32B	0.96	0.725	0.661	0.795	0.462	**0.543**	0.735	0.850
Qwen2.5-32B-Instruct	0.98	**0.674**	**0.639**	**0.726**	**0.409**	0.579	**0.645**	**0.880**
OLMo-3-1025-7B	0.81	**0.794**	0.687	**0.837**	**0.524**	**0.522**	**0.747**	**0.719**
OLMo-3-7B-Instruct	0.95	0.823	**0.659**	0.882	0.558	0.671	0.779	0.577
OLMo-3-1125-32B	0.89	0.744	0.662	0.812	0.481	**0.532**	0.706	0.839
OLMo-3.1-32B-Instruct	0.99	**0.701**	**0.644**	**0.774**	**0.436**	0.583	**0.634**	**0.853**

Table 12: Robustness of the stated-list result (DDXPlus, T=0.7, 980 cases, valid samples). Complete lists: share naming five on-list conditions with valid probabilities. TV of one list on complete lists only, and on samples whose top answer is right or wrong. Excess TV: TV minus the expected TV of the matching oracle (the reference's top five for one list; as many reference draws as the model has valid samples for the samples). Severe named, ref. < 0.1: share of named severe conditions that the reference gives less than 0.1 (over-naming). pass@32 list-sampled: top answers drawn from each sample's stated list. Bold: better model of each pair. Excess TV is averaged over samples against each case's floor, so it is not exactly the difference of the mean TV and the mean floor. A uniform guesser scores about 0.48 at pass@32.

Control	(a) View kept, variation removed	(b) Samples collapse	(c) First-answer gain is sampling noise	(d) Stated list improves (Qwen)	(e, f) Severe conditions; must-not-miss prompt	(g) Voting and agreement weaken
Worked examples, reference lists (§F.1)	holds	holds, larger (15–30 pts)	–	holds for Qwen3-8B, Qwen2.5-7B; weaker at 32B	partly: mean falls for Qwen2.5-32B, OLMo 3 7B	holds
Worked examples from another model (§F.2)	–	holds (13–30 pts)	–	holds for Qwen and Mistral, not OLMo	no excess losses (count test)	holds
Matched temperature (§F.2)	–	Qwen3-8B holds; Qwen2.5-7B-Instruct matches at T=1.3	–	holds	–	same split as (b)
Greedy decoding, matched prompts (§F.3)	–	–	holds in 4 of 6 pairs; not Qwen3-8B, nor Mistral with shared examples	holds for Qwen; OLMo 3 7B farther	–	–
Temperature frontier (§F.3)	–	holds for Qwen3-8B, OLMo 3 7B; curves meet for Qwen2.5-7B	–	list unchanged across T	–	agreement below base at every T
Must-not-miss prompt (§F.3)	–	–	–	–	coverage +6–15 pts, partly by length	–
Published must-not-miss sets (§F.4)	–	–	–	–	mean up in 7 of 8; specific losses only OLMo, Mistral, few-shot Qwen2.5	–
Format, accuracy, floor, complete lists (§E.2)	–	–	–	holds	–	–
Other families' base models (Table 24)	holds across families; blurs within	–	–	–	–	–
32B pairs (§D.5)	holds	n.s. zero-shot; holds with examples	holds (both pairs)	holds for Qwen; partly OLMo	holds for Qwen; OLMo n.s.	partly: agreement drop n.s. for OLMo
Mistral-7B, third family (§F.5)	holds	holds	fails with shared examples	holds, smallest gain	no mean gain; 3 of 17 lost significantly	holds
Thinking mode (§G.1)	holds	fails: pass@32 restored, still narrow	–	holds, closest list	–	fails: agreement better again
Rare-disease cases (§H.3)	–	–	–	–	–	holds; agreement stays the better signal

Table 13: Which claims each control supports. "Holds": the control reproduces the claim; "partly": in some pairs or settings; "fails": it reverses or disappears; –: not tested. Claims are lettered as in Table 3, which lists the evidence claim by claim;

Model	Invalid (%)	pass@1	pass@32	Cov@32	TV, one list all	TV, one list w/o true	DDF1	TV samples	Vote@32 gain	Top-3 list − samples	AUROC stated	AUROC agree@5	AUROC agree@32
Qwen3-8B-Base	0.1	0.403	**0.809**	**0.383**	0.768	0.860	0.245	**0.732**	+8.4	−16.0	0.722	0.828	0.862
Qwen3-8B	0.0	**0.601**	0.639	0.221	**0.682**	**0.805**	**0.327**	0.786	+0.2	+14.5	0.693	0.563	0.584
Qwen2.5-7B (base)	0.4	0.398	**0.761**	**0.333**	0.784	0.879	0.238	**0.756**	+6.2	−15.6	0.734	0.792	0.827
Qwen2.5-7B-Instruct	0.0	**0.450**	0.535	0.220	**0.702**	**0.806**	**0.309**	0.798	+0.6	+16.7	0.757	0.594	0.626
Qwen2.5-32B (base)	0.1	0.647	**0.896**	**0.343**	0.712	0.847	0.298	**0.715**	+7.1	−8.2	0.726	0.838	0.875
Qwen2.5-32B-Instruct	0.0	**0.663**	0.744	0.252	**0.654**	**0.775**	**0.349**	0.758	+0.6	+10.1	0.718	0.647	0.694
OLMo-3-1025-7B (base)	0.2	0.306	**0.857**	**0.454**	**0.787**	**0.859**	**0.245**	**0.712**	+10.9	−22.0	0.674	0.823	0.871
OLMo-3-7B-Instruct	0.1	**0.338**	0.558	0.213	0.808	0.884	0.238	0.830	+0.7	+0.2	0.733	0.668	0.696
OLMo-3-1125-32B (base)	0.4	0.528	**0.888**	**0.397**	0.719	0.818	0.302	**0.699**	+8.9	−13.4	0.670	0.826	0.865
OLMo-3.1-32B-Instruct	0.0	**0.575**	0.697	0.249	**0.697**	**0.812**	**0.311**	0.772	+1.1	+3.7	0.772	0.684	0.744

Table 14: Few-shot elicitation on DDXPlus (T=0.7, 980 cases, 32 samples; three worked examples in the prompt for every model, §F.1). Invalid: samples naming no listed condition. TV: distance to the reference differential, for one stated list (with the true pathology removed from both in "w/o true") and for the sampled top answers. Vote@32 gain and top-3 list minus top-3 of the samples are in accuracy points. AUROC: for the single-call answer being correct, of its stated probability and of agreement with the next 4 or the next 31 samples. Bold: better of the pair.

Pair (base → post)	pass@32 (points)	TV samples	TV, one list all	TV, one list w/o true	DDF1	Reference mass of one list
Qwen3-8B	−12.4 / −17.0	+0.043 / +0.054	−0.043 / −0.087	−0.018† / −0.055	+0.042 / +0.081	+0.062 / +0.113
Qwen2.5-7B	−9.7 / −22.7	+0.028† / +0.042	−0.062 / −0.082	−0.041 / −0.073	+0.062 / +0.072	+0.073 / +0.124
Qwen2.5-32B	−4.9† / −15.2	+0.033 / +0.044	−0.053 / −0.057	−0.043 / −0.072	+0.051 / +0.051	+0.067 / +0.084
OLMo 3 7B	−28.3 / −29.9	+0.090 / +0.118	+0.029† / +0.021†	+0.034 / +0.024	−0.006† / −0.007†	−0.010† / −0.002†
OLMo 3.1 32B	−4.0† / −19.1	+0.020† / +0.074	−0.045 / −0.022†	−0.023 / −0.006†	+0.040 / +0.009†	+0.056 / +0.047

Table 15: Base → post-trained differences under the zero-shot prompt / the few-shot prompt (DDXPlus, T=0.7). †: not significant under the condition-clustered bootstrap with Holm correction (over 112 tests for the zero-shot and 70 for the few-shot pairs; Appendix J).

Pair (base → post)	pass@32 (points)	TV samples	TV, one list all	TV, one list w/o true	DDF1	Cond. TV
Qwen3-8B	−14.6	+0.049	−0.077	−0.049	+0.079	−0.078
Qwen2.5-7B	−19.7	+0.031†	−0.077	−0.059	+0.074	−0.094
Qwen2.5-32B	−14.5	+0.036	−0.053	−0.057	+0.054	−0.051
OLMo 3 7B	−29.7	+0.116	−0.007†	+0.013†	−0.002†	−0.015†
OLMo 3.1 32B	−17.4	+0.073	−0.016†	+0.002†	+0.009†	−0.005†
Mistral-7B	−12.7	+0.038†	−0.059	−0.031	+0.057	−0.100

Table 16: Few-shot control with reference-free examples (DDXPlus, T=0.7, 980 cases): the same three training patients as in §F.1, but each example list is Qwen2.5-32B-Instruct's own greedy answer rather than the reference's top five. Post-trained minus base. †: not significant (condition-clustered bootstrap, Holm over 84 tests; conditional TV Holm-corrected as a separate family).

Model	T	Invalid (%)	pass@1	pass@32	Distinct	TV samples	TV one list	Vote@32 gain	AUROC stated	AUROC agree@32
Qwen3-8B-Base	0.7	11.3	0.388	0.732	3.77	0.750	0.743	+3.4	0.595	0.810
	1	22.3	0.282	0.789	6.60	0.715	0.775	+10.4	0.597	0.857
Qwen3-8B	0.7	0.0	0.559	0.607	1.27	0.793	0.700	+0.3	0.734	0.655
	1	0.0	0.556	0.643	1.53	0.783	0.700	+1.4	0.726	0.701
	1.3	0.0	0.552	0.668	1.73	0.776	0.701	+1.0	0.707	0.724
	1.6	0.0	0.549	0.684	1.92	0.770	0.701	+1.8	0.715	0.745
Qwen2.5-7B	0.7	0.9	0.352	0.695	4.91	0.757	0.779	+5.2	0.576	0.811
	1	18.4	0.239	0.766	8.23	0.719	0.805	+14.1	0.587	0.867
Qwen2.5-7B-Instruct	0.7	0.0	0.478	0.598	1.82	0.785	0.717	+0.8	0.571	0.702
	1	0.0	0.464	0.663	2.65	0.761	0.717	+4.4	0.583	0.767
	1.3	2.2	0.436	0.704	3.38	0.745	0.722	+5.6	0.602	0.794
	1.6	30.5	0.304	0.726	3.88	0.729	0.749	+10.0	0.637	0.814

Table 17: Matched-temperature control (DDXPlus, 980 cases, 32 samples, zero-shot prompt): the post-trained models sampled at higher temperatures (top-p 1) against their base models at T=0.7 and T=1. Columns as in Tables 1, 2 and 26.

	Top-1 accuracy				Top-1, post − base (points)		
	base, examples		post-trained		same	ref.-free	
Pair	reference	ref.-free	zero-shot	examples	examples	base	original
Qwen3-8B	0.471	0.450	0.558	0.607	+13.7	+10.7	+8.7†
Qwen2.5-7B	0.437	0.420	0.496	0.458	+1.8†	+7.6†	+5.7†
Qwen2.5-32B	0.712	0.686	0.606	0.664	−4.8†	−8.0	−10.6
OLMo 3 7B	0.403	0.403	0.258	0.345	−5.8†	−14.5	−14.5
OLMo 3.1 32B	0.630	0.559	0.558	0.571	−5.7†	−0.1†	−7.1†
Mistral-7B	0.468	0.446	0.417	0.577	+10.8	−2.9†	−5.1†

	Recall@5, post − base (points)			TV one list, post − base			Severe named, post − base		
Pair	same	ref.-free	original	same	ref.-free	original	same	ref.-free	original
Qwen3-8B	+26.0	+21.6	+24.2	−0.108	−0.077	−0.088	+0.127	+0.127	+0.127
Qwen2.5-7B	+20.1	+18.3	+16.8	−0.074	−0.075	−0.060	+0.091	+0.119	+0.110
Qwen2.5-32B	+0.3†	−2.4†	−5.3†	−0.054	−0.035	−0.031	+0.058	+0.046	+0.062
OLMo 3 7B	+2.0†	−17.2	−14.1	+0.060	+0.067	+0.083	+0.004†	−0.058†	−0.020†
OLMo 3.1 32B	+2.5†	−0.5†	−2.0†	+0.004†	+0.004†	+0.012†	−0.005†	−0.033†	−0.042†
Mistral-7B	+16.7	+2.9†	+4.1†	−0.009†	+0.008†	+0.029	−0.034†	−0.034†	−0.059†

Table 18: Greedy single call (DDXPlus, 980 cases, temperature 0, one answer per case). Base models are prompted with the worked examples of §F.1, whose lists are the reference top five ("reference") or Qwen2.5-32B-Instruct's answers ("ref.-free"); post-trained models are prompted zero-shot or with the reference examples. Three comparisons: *same examples* (both arms see the reference examples), *ref.-free base* (reference-free base against zero-shot post-trained, so neither arm sees a reference list), and *original* (reference-example base against zero-shot post-trained). Differences are over cases where both models gave a valid answer, so they can differ slightly from the difference of the level columns. †: the condition-clustered 95% interval includes zero.

Model	T	Invalid (%)	pass@1	pass@32	Distinct	TV one list	AUROC stated	AUROC agree@32
Qwen3-8B-Base	0.4	0.1	0.447	0.727	3.64	0.772	0.687	0.806
	0.7	—	0.388	0.832	7.09	0.774	0.708	0.876
	1.0	—	0.304	0.880	10.33	0.800	0.703	0.909
	1.3	29.5	0.179	0.857	10.39	0.850	0.630	0.921
Qwen3-8B	0.4	0.0	0.602	0.632	1.14	0.681	0.693	0.569
	0.7	—	0.601	0.655	1.25	0.682	0.693	0.613
	1.0	—	0.599	0.676	1.40	0.682	0.695	0.651
	1.3	0.0	0.596	0.703	1.57	0.682	0.694	0.679
Qwen2.5-7B	0.4	0.4	0.421	0.662	3.13	0.778	0.748	0.805
	0.7	—	0.388	0.784	5.72	0.790	0.719	0.843
	1.0	—	0.333	0.841	8.53	0.815	0.692	0.899
	1.3	25.3	0.220	0.848	9.24	0.857	0.615	0.916
Qwen2.5-7B-Instruct	0.4	0.0	0.455	0.516	1.40	0.702	0.750	0.613
	0.7	—	0.448	0.556	1.78	0.703	0.747	0.643
	1.0	—	0.441	0.597	2.28	0.704	0.741	0.687
	1.3	0.2	0.430	0.660	2.98	0.707	0.720	0.729
OLMo-3-1025-7B	0.4	0.1	0.368	0.803	5.46	0.762	0.667	0.815
	0.7	—	0.294	0.865	9.62	0.795	0.663	0.879
	1.0	—	0.214	0.881	12.40	0.834	0.618	0.899
	1.3	36.2	0.110	0.801	10.69	0.888	0.552	0.882
OLMo-3-7B-Instruct	0.4	0.1	0.345	0.504	1.99	0.809	0.737	0.663
	0.7	—	0.335	0.602	2.95	0.808	0.740	0.716
	1.0	—	0.324	0.668	4.09	0.809	0.744	0.760
	1.3	5.6	0.286	0.737	5.88	0.831	0.680	0.775

Table 19: Temperature frontier with the few-shot prompt (DDXPlus, 980 cases, 32 samples, top-p 1 at every temperature). Invalid rates were computed for T=0.4 and 1.3 only.

Model	Valid (%)	Conditions likely + mnm	Recall@5 default	Recall@5 likely part	Severe named (ref. ≥ 0.1) base	Severe named (ref. ≥ 0.1) default	Severe named (ref. ≥ 0.1) mnm	Unwarranted severe default	Unwarranted severe mnm	Acute pulm. edema default	Acute pulm. edema mnm
Qwen3-8B	100	4.1 + 2.3	0.802	0.715	0.318	0.437	0.525	1.06	2.11	0.61	0.60
Qwen2.5-7B-Instruct	99	3.9 + 2.9	0.735	0.640	0.304	0.439	0.503	1.57	2.95	0.63	0.71
Qwen2.5-32B-Instruct	100	4.8 + 2.9	0.840	0.838	0.414	0.472	0.579	0.91	2.27	0.44	0.63
OLMo-3-7B-Instruct	96	3.7 + 2.5	0.485	0.501	0.332	0.317	0.444	1.15	2.60	0.31	0.58
OLMo-3.1-32B-Instruct	100	4.9 + 3.0	0.779	0.790	0.376	0.392	0.545	0.74	2.85	0.41	0.59
Mistral-7B-Instruct	87	3.7 + 2.7	0.613	0.562	0.352	0.329	0.473	0.86	2.29	0.59	0.68

Table 20: Must-not-miss prompt (DDXPlus, T=0.7, 980 cases, 32 samples): the five most likely conditions, then up to three dangerous conditions to rule out. *Default*: the same model with the standard zero-shot prompt; *base*: its base model with the few-shot prompt. Recall@5 of the most-likely part; severe conditions and unwarranted severe conditions (severe, reference < 0.1, per list) over the whole list; acute pulmonary edema: share of lists naming it where the reference gives it any weight.

Pair	Union set (7 conditions) base	post	post − base	sig. lost / gained	Harm-based post − base	Chest pain post − base
Qwen3-8B, T=0.7	0.494	0.584	+0.089	0 / 2	+0.053	+0.115
Qwen3-8B, T=1	0.425	0.585	+0.160	0 / 4	+0.135	+0.167
Qwen2.5-7B, T=0.7	0.315	0.536	+0.221	0 / 4	+0.267	+0.149
Qwen2.5-7B, T=1	0.303	0.539	+0.236	0 / 4	+0.287	+0.180
Qwen2.5-32B, T=0.7	0.432	0.595	+0.162	0 / 4	+0.164	+0.104
OLMo 3 7B, T=0.7	0.268	0.413	+0.144	2 / 3	+0.133	+0.101
OLMo 3 7B, T=1	0.275	0.412	+0.136	3 / 3	+0.135	+0.102
OLMo 3.1 32B, T=0.7	0.462	0.459	−0.003†	3 / 0	−0.033†	−0.020†
Qwen3-8B, few-shot	0.299	0.491	+0.191	0 / 3	+0.183	+0.093
Qwen2.5-7B, few-shot	0.239	0.455	+0.216	2 / 3	+0.249	+0.152
Qwen2.5-32B, few-shot	0.361	0.542	+0.180	1 / 3	+0.210	+0.096
OLMo 3 7B, few-shot	0.335	0.409	+0.074	2 / 3	−0.015†	+0.036†
OLMo 3.1 32B, few-shot	0.412	0.477	+0.065†	2 / 1	+0.080	−0.017†
Mistral-7B, few-shot	0.401	0.441	+0.040†	4 / 1	+0.024†	−0.071

Post-trained model	Warranted must-not-miss conditions named (union set) standard	two lists	must-not-miss	− standard	Unwarranted per list standard	two lists	must-not-miss
Qwen3-8B	0.584	0.612	0.775	+0.192	1.08	1.20	2.10
Qwen2.5-7B-Instruct	0.536	0.600	0.714	+0.178	0.94	1.14	2.33
Qwen2.5-32B-Instruct	0.595	0.648	0.814	+0.219	1.00	1.21	2.47
OLMo-3-7B-Instruct	0.413	0.479	0.564	+0.151	0.93	1.17	1.60
OLMo-3.1-32B-Instruct	0.459	0.527	0.715	+0.256	0.58	0.77	1.77
Mistral-7B-Instruct	0.358	0.417	0.615	+0.257	0.69	0.87	1.65

Table 21: Must-not-miss conditions defined outside DDXPlus (DDXPlus, 980 cases). *Harm-based*: the DDXPlus conditions among the 15 diseases behind about half of serious misdiagnosis-related harms in the USA (Newman-Toker et al., 2024) (NSTEMI/STEMI, pulmonary embolism, pneumonia, pulmonary neoplasm). *Chest pain*: the life-threatening causes of acute chest pain in the 2021 AHA/ACC guideline (Gulati et al., 2021) (NSTEMI/STEMI, unstable angina, pulmonary embolism, Boerhaave, spontaneous pneumothorax). *Union*: either set. Top: share of the set's conditions with reference probability ≥ 0.1 that one list names, and the number of the union's conditions named significantly less / more often (Benjamini–Hochberg over the 7 conditions). Bottom: the must-not-miss prompt (§F.3) scored against the union set; *two lists*: the union of two standard lists, a length-matched comparison; unwarranted: the set's conditions named that the reference weights below 0.1. †: the condition-clustered 95% interval includes zero.

Model	Invalid (%)	pass@1	pass@32	Distinct	Cov@32	TV, one list all	TV, one list w/o true	DDF1	TV samples	Vote@32 gain	Top-3 list − samples	AUROC stated	AUROC agree@5
Mistral-7B-v0.3 (base), few-shot	0.4	0.428	**0.791**	**5.02**	**0.355**	0.777	0.875	0.250	**0.741**	+5.6	−12.6	0.671	0.826
Mistral-7B-Instruct-v0.3, few-shot	0.3	**0.568**	0.646	1.34	0.217	**0.741**	**0.851**	**0.296**	0.799	+0.8	+7.0	0.747	0.613
Qwen3-8B-Base	11.3	0.388	0.732	**3.77**	**0.323**	0.743	0.833	0.276	0.750	+3.4	−7.3	0.595	0.772
Qwen3-8B, non-thinking	0.0	0.559	0.607	1.27	0.217	0.700	0.815	0.318	0.793	+0.3	+13.7	0.734	0.613
Qwen3-8B, thinking	0.0	**0.671**	**0.815**	1.76	0.283	**0.670**	**0.799**	**0.349**	**0.745**	+1.5	+3.0	0.638	0.693

Table 22: A third family and a reasoning mode on DDXPlus (T=0.7, 980 cases, 32 samples). Top: Mistral-7B v0.3 with the few-shot prompt of §F.1. Bottom: Qwen3-8B-Base and Qwen3-8B with thinking off (the main runs) and on, with the zero-shot prompt. Columns as in Table 14; Distinct: distinct top answers per 32 samples. Bold: best within each block.

Pair	TV, base model 1 list	TV, base model mean of 5	TV, base model mean of 32	TV, post-trained model 1 list	TV, post-trained model mean of 5	TV, post-trained model mean of 32	Post − base, 5 lists each TV	Post − base, 5 lists each Severe named
Zero-shot prompt								
Qwen3-8B	0.743	0.699	0.691	0.700	0.695	0.692	−0.004†	+0.030†
Qwen2.5-7B	0.779	0.717	0.706	0.717	0.692	0.685	−0.025	+0.023†
Qwen2.5-32B	0.726	0.681	0.674	0.674	0.658	0.654	−0.024	−0.037†
OLMo 3 7B	0.794	0.730	0.717	0.823	0.805	0.801	+0.074	−0.076
OLMo 3.1 32B	0.746	0.691	0.678	0.701	0.677	0.673	−0.014†	−0.088
Few-shot prompt								
Qwen3-8B	0.768	0.704	0.689	0.682	0.673	0.672	−0.030	−0.006†
Qwen2.5-7B	0.784	0.712	0.694	0.702	0.686	0.683	−0.027	+0.012†
Qwen2.5-32B	0.712	0.655	0.641	0.654	0.640	0.638	−0.015†	−0.094
OLMo 3 7B	0.787	0.703	0.676	0.808	0.786	0.778	+0.083	−0.141
OLMo 3.1 32B	0.719	0.651	0.630	0.697	0.682	0.680	+0.031	−0.060
Mistral-7B v0.3	0.777	0.708	0.691	0.741	0.723	0.720	+0.015†	−0.118
Zero-shot prompt, thinking on								
Qwen3-8B, thinking	0.743	0.699	0.691	0.670	0.650	0.647	−0.049	+0.129

Table 23: Averaging stated lists across calls (DDXPlus, T=0.7, 980 cases). TV to the reference differential of one stated list (per-case mean over valid lists, as in Table 2) and of the mean of the first 5 or 32 valid lists, for the base and the final post-trained model. Last two columns: post-trained minus base with five averaged lists each, for TV and for the share of severe conditions with reference probability ≥ 0.1 that are named. †: the condition-clustered 95% interval includes zero.

Pair	TV of a post-trained list to own base mean	TV of a post-trained list to same-diagnosis means	reference	Within-model benchmark base list	Within-model benchmark post list	Differences own − same dx	Differences own − base list
Zero-shot prompt							
Qwen3-8B, T=0.7	0.368	0.444	0.701	0.372	0.109	−0.076	−0.004†
Qwen3-8B, T=1	0.424	0.486	0.701	0.551	0.153	−0.062	−0.128
Qwen2.5-7B, T=0.7	0.464	0.519	0.717	0.470	0.248	−0.055	−0.006†
Qwen2.5-7B, T=1	0.532	0.575	0.717	0.650	0.315	−0.043	−0.119
Qwen2.5-32B, T=0.7	0.360	0.431	0.674	0.375	0.189	−0.071	−0.015†
OLMo 3 7B, T=0.7	0.628	0.658	0.823	0.493	0.238	−0.031	+0.135
OLMo 3 7B, T=1	0.651	0.679	0.818	0.691	0.326	−0.028	−0.040
OLMo 3.1 32B, T=0.7	0.386	0.452	0.700	0.432	0.227	−0.066	−0.046
Few-shot prompt							
Qwen3-8B	0.469	0.518	0.681	0.471	0.102	−0.049	−0.003†
Qwen2.5-7B	0.559	0.596	0.702	0.515	0.171	−0.037	+0.044
Qwen2.5-32B	0.465	0.511	0.654	0.441	0.154	−0.045	+0.024
OLMo 3 7B	0.669	0.710	0.808	0.594	0.259	−0.040	+0.075
OLMo 3.1 32B	0.482	0.532	0.697	0.469	0.150	−0.050	+0.014†
Mistral-7B	0.495	0.546	0.740	0.495	0.181	−0.051	−0.000†
Thinking on							
Qwen3-8B, thinking	0.412	0.468	0.669	0.372	0.189	−0.057	+0.040

Table 24: Does a post-trained list reproduce its base model's view of the patient? (DDXPlus, 980 cases.) TV between a post-trained list and the mean of the base model's valid lists for the same case (own base mean), the base model's means for the other cases with the same true pathology, and the reference. Within-model benchmarks: a base list against the mean of the base model's other lists for the case, and a post-trained list against the mean of the post-trained model's other lists. †: the condition-clustered 95% interval includes zero.

Model (T=0.7)	Stated p	Agree@5	Agree@32
Qwen3-8B-Base	+0.042	+0.005	+0.024
Qwen3-8B	+0.116	+0.056	+0.098
Qwen2.5-7B	+0.044	+0.025	+0.038
Qwen2.5-7B-Instruct	+0.070	+0.054	+0.070
Qwen2.5-32B	+0.064	−0.026	−0.038
Qwen2.5-32B-Instruct	−0.043	−0.060	−0.109
OLMo-3-1025-7B	+0.083	−0.048	−0.052
OLMo-3-7B-Instruct	−0.067	−0.073	−0.049
OLMo-3-1125-32B	−0.005	+0.020	+0.011
OLMo-3.1-32B-Instruct	+0.063	+0.014	−0.023

Table 25: Spearman correlation between each uncertainty signal (lower confidence or agreement) and the entropy of the case's reference differential (DDXPlus, 980 cases). Neither readout tracks how ambiguous a case is.

Model	Single-call accuracy	Vote@32 gain	Top-3: list − 32 samples	AUROC, single-call answer correct		
				Stated p	Agree@5	Agree@32
Qwen3-8B-Base	0.456	+3.4	−7.3	0.595	0.772	**0.810**
Qwen3-8B	0.556	+0.3	+13.7	**0.734**	0.613	0.655
Qwen3-8B (CoT prompt)	0.648	+4.5	+0.0	0.718	0.725	**0.778**
Qwen3-8B (thinking)	0.672	+1.5	+3.0	0.638	0.693	**0.732**
Qwen2.5-7B	0.371	+5.2	−8.9	0.576	0.764	**0.811**
Qwen2.5-7B-Instruct	0.480	+0.8	+7.3	0.571	0.653	**0.702**
Qwen2.5-32B	0.510	+7.0	−8.3	0.730	0.759	**0.797**
Qwen2.5-32B-Instruct	0.598	+1.9	+5.6	0.709	0.663	**0.721**
OLMo-3 7B base	0.291	+4.0	−13.6	0.538	0.761	**0.802**
OLMo-3-7B-Instruct	0.267	−0.2	+1.9	**0.796**	0.668	0.731
OLMo-3.1 32B base	0.441	+8.4	−13.2	0.629	0.805	**0.831**
OLMo-3.1-32B-Instruct	0.562	+0.9	−0.6	0.696	0.712	**0.772**
AlphaMed-7B	0.528	+0.9	+0.6	0.673	0.663	**0.715**
HuatuoGPT-o1-7B	0.532	+4.8	−0.4	0.590	0.715	**0.775**
m1-7B	0.495	+3.8	−4.6	0.680	0.736	**0.782**

Table 26: Aggregation and uncertainty on DDXPlus (T=0.7, 980 cases). Single-call accuracy: first valid sample. Vote@32 gain: majority answer over valid samples minus single call (points). Top-3: one list's top three minus the three most frequent sampled answers, containing the true pathology (points). AUROC for the single-call answer being correct; bold marks the best signal.

Model	Post-training	pass@1	pass@32	Distinct	Entropy	Cov@32	ECE
Qwen2.5-7B-Instruct	(parent)	0.478	0.598	1.82	0.30	0.239	**0.106**
AlphaMed-7B	rule-based RL	0.516	0.682	1.87	0.28	0.266	0.121
HuatuoGPT-o1-7B	SFT + PPO	**0.536**	0.735	2.49	0.44	0.306	0.168
m1-7B-23K	SFT distillation	0.486	0.779	3.29	0.62	0.328	0.206
Qwen2.5-7B (base)	(parent)	0.352	0.695	**4.91**	**1.00**	**0.335**	0.291
AlphaMed-7B (base-RL)	rule-based RL	0.495	**0.790**	3.14	0.62	0.320	0.202

Table 27: Medical models on DDXPlus (980 cases, 32 samples, T=0.7). Upper block: descendants of Qwen2.5-7B-Instruct; none narrows the differential further, and the reasoning-trained models widen it. Lower block: AlphaMed's companion checkpoint trained from the base model, which keeps most of the base model's diversity and has the best zero-shot pass@32 of any 7B model.

Model	≤2022 (n=901) pass@1	≤2022 (n=901) pass@32	2023–24 (n=348) pass@1	2023–24 (n=348) pass@32	2025+ (n=152) pass@1	2025+ (n=152) pass@32
Qwen2.5-7B (base)	2.0	8.2	0.0	0.3	0.6	3.3
Qwen2.5-7B-Instruct	2.3	7.7	0.0	0.9	0.8	3.9
AlphaMed-7B	3.0	9.0	0.3	1.4	1.4	3.9
AlphaMed-7B (base-RL)	2.4	8.8	0.1	1.1	1.4	4.6
HuatuoGPT-o1-7B	3.1	8.4	0.3	1.4	1.3	3.9
m1-7B	2.6	9.5	0.3	1.7	1.0	3.9
Qwen3-8B-Base	2.1	8.3	0.1	0.6	1.1	3.9
Qwen3-8B	3.1	5.2	0.1	0.3	2.0	2.6

Table 28: Phenopacket Store accuracy (%, exact diagnosis) by publication year of the source case report (1,401 cases, 32 samples, T=0.7). Every model is far more accurate on cases published up to 2022 than on 2023–24 cases; 2025+ cases fall in between.

Model	Single-call correct	Vote@32 gain	Top-3: list − 32 samples	AUROC, single-call answer correct: Stated p	Agree@5	Agree@32	Correct answers
Qwen3-8B-Base	0.074	+4.6	−7.0	0.567	0.768	**0.830**	104
Qwen3-8B	0.080	+0.4	+0.4	0.707	0.726	**0.763**	112
Qwen2.5-7B	0.062	+3.8	−5.4	0.598	0.788	**0.845**	87
Qwen2.5-7B-Instruct	0.056	+1.5	−3.6	0.717	0.661	**0.745**	78
OLMo-3-1025-7B	0.060	+3.1	−4.6	0.464	0.765	**0.814**	84
SFT	0.035	+1.1	−3.6	0.628	0.730	**0.814**	49
SFT+DPO	0.026	+1.2	−3.3	0.651	0.673	**0.725**	36
SFT+DPO+RLVR (Instruct)	0.035	+0.9	−2.7	0.652	0.644	**0.726**	49
RL-Zero (from base)	0.065	+2.3	−3.1	0.583	0.791	**0.827**	91
AlphaMed-7B	0.071	+1.6	−3.0	0.586	0.755	**0.797**	100
HuatuoGPT-o1-7B	0.093	+2.6	−5.7	0.653	0.708	**0.756**	130
m1-7B	0.084	+2.9	−7.9	0.650	0.747	**0.800**	117
AlphaMed-7B (base-RL)	0.054	+2.5	−4.2	0.672	0.733	**0.789**	75

Table 29: Aggregation and uncertainty on Phenopacket Store (1,401 real rare-disease cases, 32 samples, T=0.7). An answer counts as correct if it is the diagnosis or a closely related Mondo class. Columns as in Table 26; Correct answers: number of correct single-call answers, the positives for the AUROCs. Bold: best signal.

Pair (post − base)	pass@32 (pts)	Distinct	TV samples	TV one list	TV list w/o gold	DDF1	AUROC agree@32
Qwen3-8B, T=0.7	−12.4 [−19.6, −6.0]	−2.50 [−2.89, −2.13]	+0.043 [+0.021, +0.064]	−0.043 [−0.059, −0.028]	−0.018† [−0.032, −0.005]	+0.042 [+0.029, +0.056]	−0.160 [−0.216, −0.096]
Qwen3-8B, T=1	−14.6 [−21.5, −8.3]	−5.06 [−5.49, −4.65]	+0.067 [+0.042, +0.091]	−0.074 [−0.090, −0.059]	−0.039 [−0.052, −0.027]	+0.066 [+0.053, +0.080]	−0.158 [−0.211, −0.104]
Qwen2.5-7B, T=0.7	−9.7 [−16.1, −4.1]	−3.09 [−3.63, −2.57]	+0.028† [+0.007, +0.049]	−0.062 [−0.076, −0.049]	−0.041 [−0.054, −0.028]	+0.062 [+0.049, +0.075]	−0.106 [−0.163, −0.044]
Qwen2.5-7B, T=1	−10.3 [−17.2, −4.2]	−5.59 [−6.03, −5.15]	+0.042† [+0.015, +0.068]	−0.088 [−0.103, −0.074]	−0.056 [−0.069, −0.043]	+0.083 [+0.069, +0.097]	−0.104 [−0.159, −0.046]
Qwen2.5-32B, T=0.7	−4.9† [−10.3, +0.5]	−2.10 [−2.49, −1.74]	+0.033 [+0.018, +0.049]	−0.053 [−0.064, −0.042]	−0.043 [−0.053, −0.032]	+0.051 [+0.041, +0.061]	−0.079 [−0.119, −0.038]
OLMo 3 7B, T=0.7	−28.3 [−35.6, −21.1]	−2.56 [−2.84, −2.27]	+0.090 [+0.071, +0.109]	+0.029† [+0.004, +0.052]	+0.034 [+0.017, +0.051]	−0.006† [−0.022, +0.009]	−0.076† [−0.157, +0.001]
OLMo 3 7B, T=1	−15.2 [−20.8, −9.7]	−3.95 [−4.22, −3.68]	+0.079 [+0.060, +0.098]	−0.009† [−0.031, +0.011]	+0.006† [−0.006, +0.018]	+0.024 [+0.010, +0.036]	−0.076† [−0.151, −0.011]
OLMo 3.1 32B, T=0.7	−4.0† [−6.8, −1.2]	−2.18 [−2.47, −1.90]	+0.020† [+0.002, +0.037]	−0.045 [−0.059, −0.031]	−0.023 [−0.034, −0.012]	+0.040 [+0.026, +0.054]	−0.056† [−0.095, −0.019]

Table 30: Base → post-trained (final Instruct) differences on DDXPlus with 95% intervals from a bootstrap over the 49 true-pathology clusters. †: not significant after Holm correction across 112 tests (50,000 resamples, add-one p-values). pass@32 here is recomputed from the samples with the analysis scripts' name matching and can differ from Table 5 by up to 0.3 points.